

PROGRESS REPORT #2

MC-ESO

Multi-Channel Epidemic Spread Optimizer

Diagnosis-driven refinements since 5/18

M1 Kosei Matsuzaki / 2026-07-14

1

Recap

Where the 5/18 report left off

Motivation & goal

Every method has a landscape it fails on

mean SR@1e-10 / BBOB-24 / dim 2

Method	Weak function shape	SR@1e-10
CMA-ES / L-SHADE	Multimodal (F15–F24)	32% / 42%
PSO / SaVOA	Ill-conditioned (F10–F14)	7% / 18%
DE	Bent-valley / asymmetric (F06–F09)	91% (rest $\approx$ 100)

GOAL

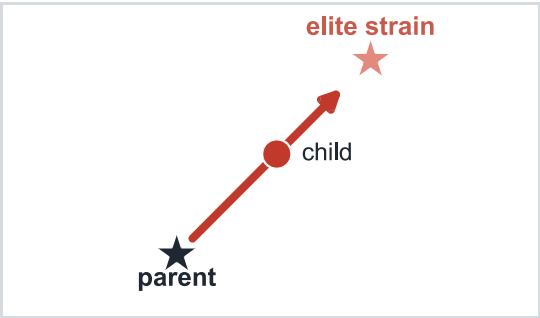
Mimic multi-route viral spread — robust across every function shape

How MC-ESO works

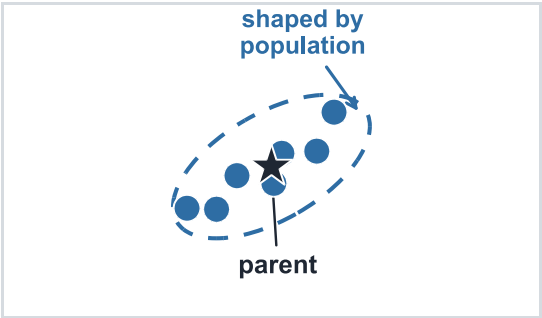
Transmission channels

three parallel reproduction routes

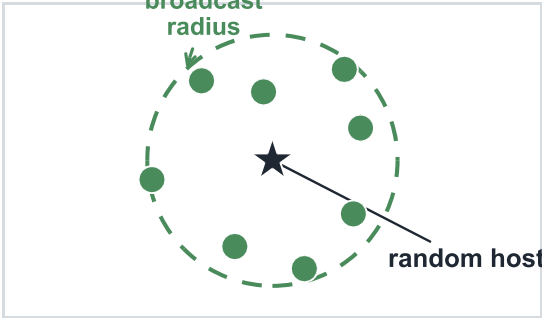
Droplet



Close-contact



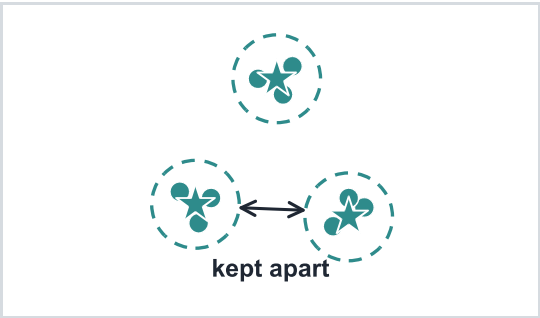
Airborne



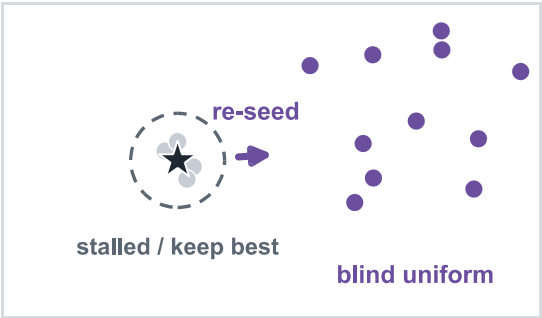
Population strategies

how the swarm is steered

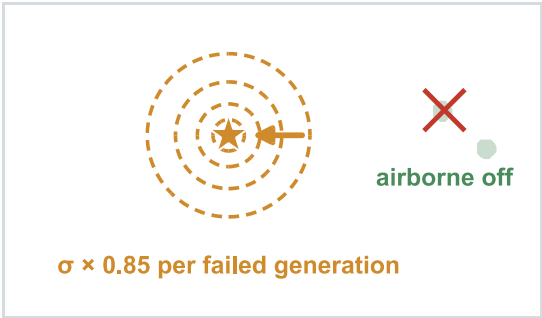
Strain coexistence



Restart (spillover)



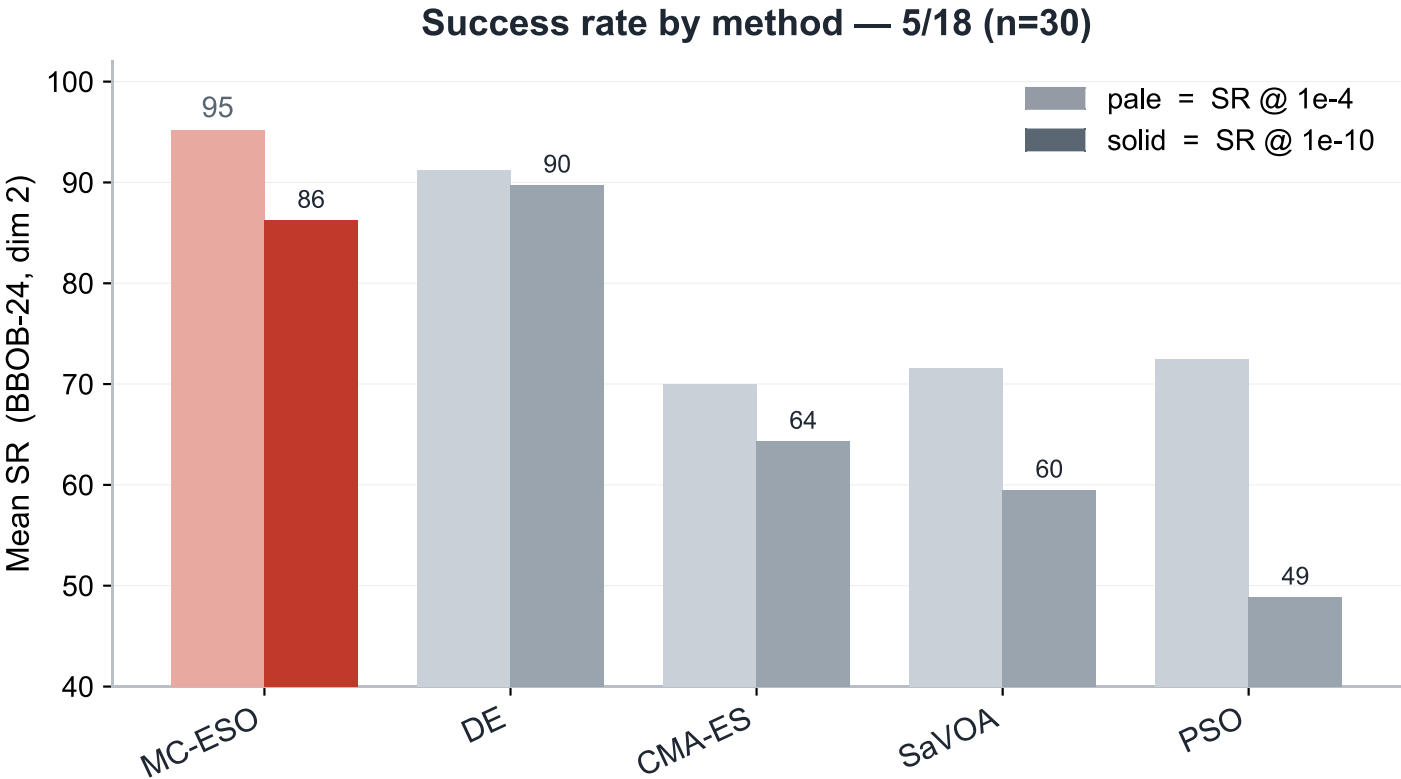
Drilling



Previous results (5/18)

Best at SR@1e-4 — but DE led at 1e-10

BBOB-24 [10] / dim 2 / n = 30



method	1e-4	1e-10
MC-ESO	95.2	86.2
DE	91.2	89.8
SaVOA	71.5	59.5
PSO	72.5	48.9
CMA-ES	70.0	64.3

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Direction

Where to grow — and what we chose

Performance first

Fix the SR@1e-10 gaps before chasing multiple optima

A. Performance

CHOSEN

- › Fill deep-precision gaps on F11–F14
- › SR@1e-10 never lowered

Primary metric = SR@1e-10

B. Multimodality

PARTIAL

- › Find many optima in parallel
- › Sequential niching — SR-safe only

Deeper work deferred → §5

3

Improvements

Four changes — what, why, and the numbers

Four changes since 5/18

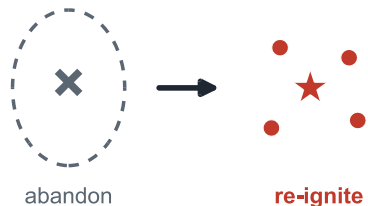
Cumulative — each builds on the last

6/12

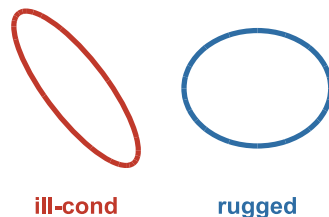
6/24

7/03

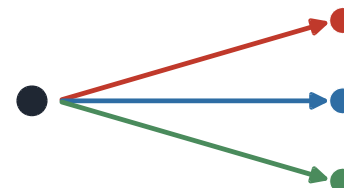
7/03



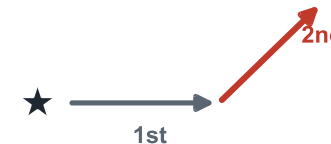
Informed restart
reservoir re-ignition + basin
repulsion



Adaptive floor
eigenvalue-ratio anisotropy



Channel router
per-landscape air-budget routing

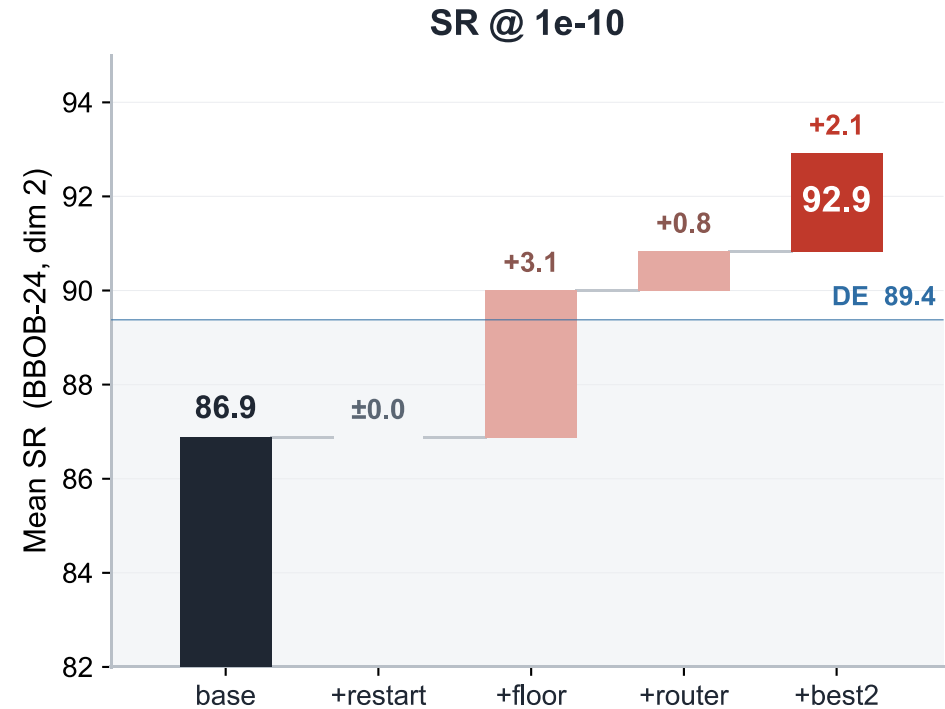
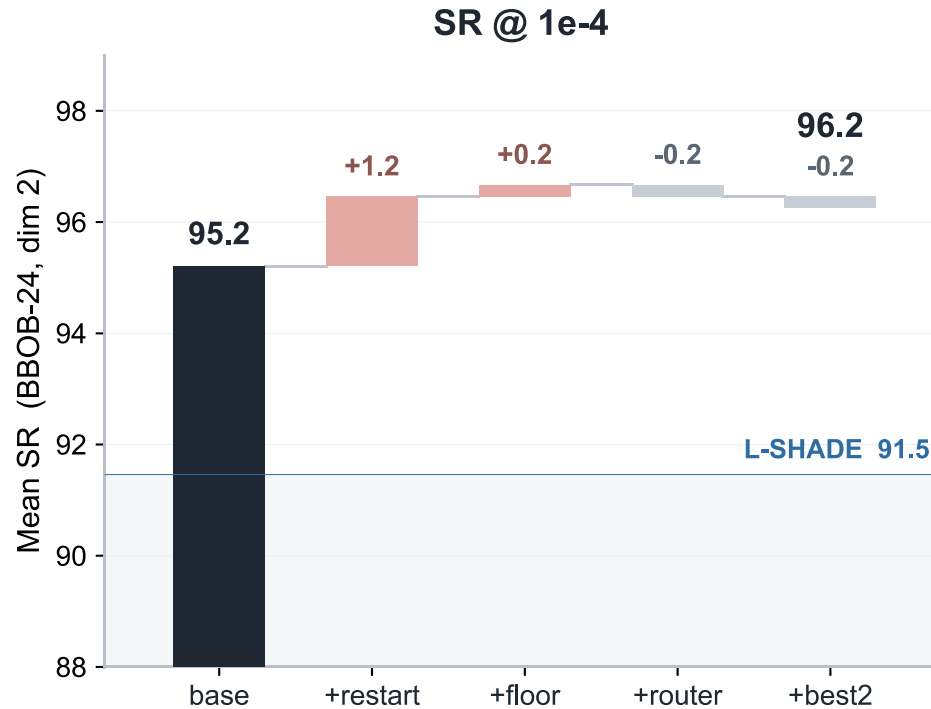


Route-gated best2
2nd difference on droplet route

Cumulative ablation — +6.0 pt

SR@1e-10 86.9 → 92.9%

BBOB-24 / dim 2 / n = 20



Now strong at both ends — coarse discovery kept (1e-4 saturated), deep precision gained (1e-10 +6, past DE)

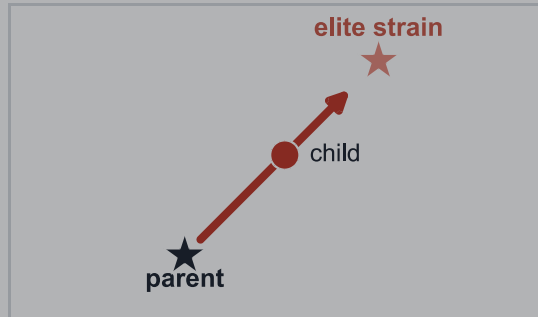
Sanity: abl0 86.9% ≈ the 86.2% measured on 5/18 (n=30)

① Informed restart

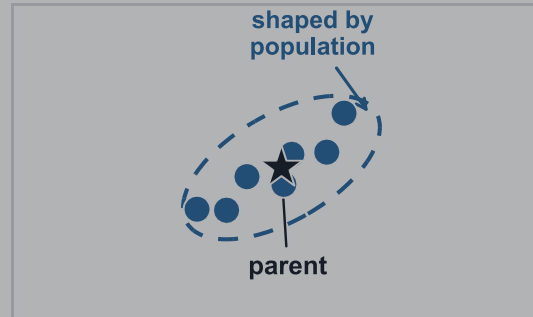
Transmission channels

three parallel reproduction routes

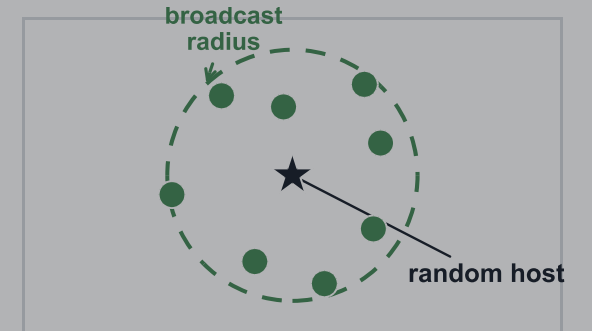
Droplet



Close-contact



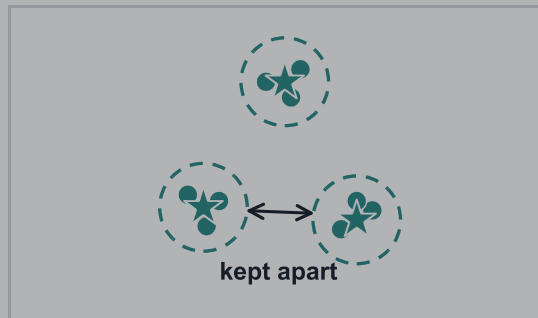
Airborne



Population strategies

how the swarm is steered

Strain coexistence



Restart (spillover)



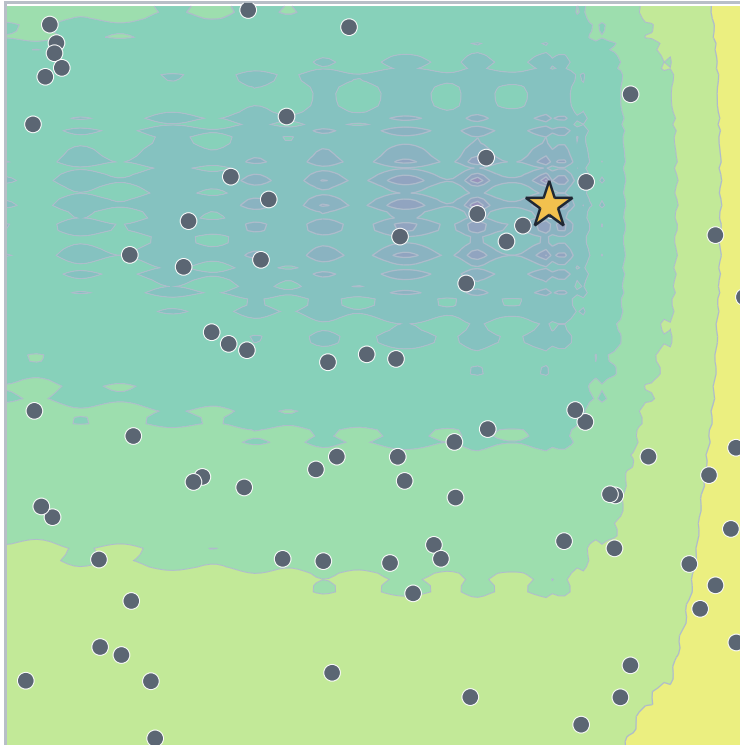
Drilling



① Informed restart

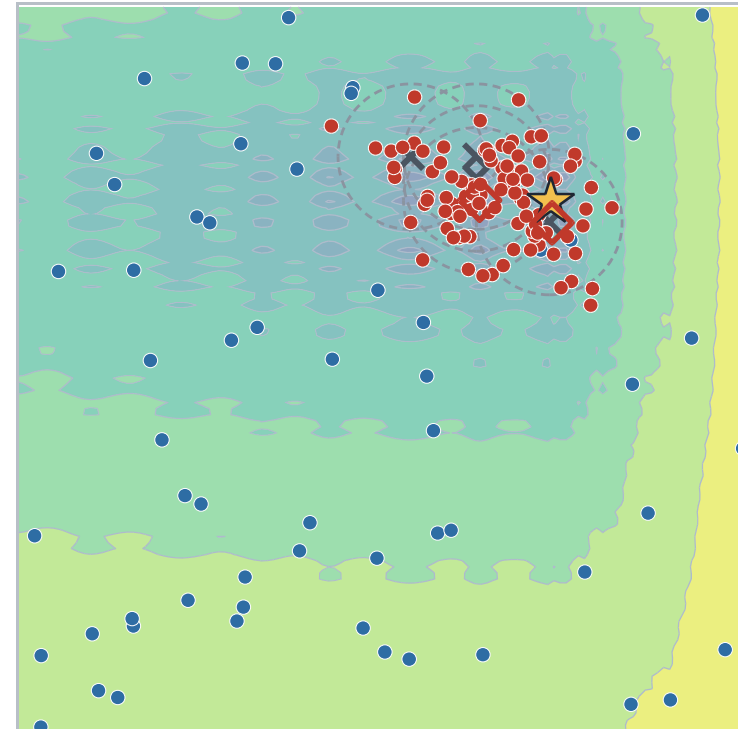
A stalled sub-population re-seeds — real run on F04

Before — blind uniform restart



re-seeds land anywhere / avg dist to optimum 4.9

After — informed restart



re-ignite good spots, avoid explored / avg dist 1.1

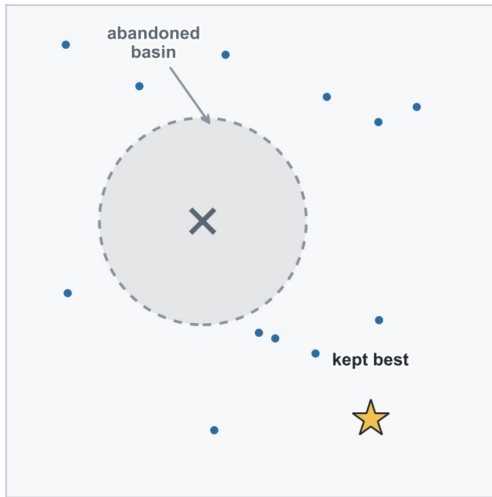
- ★ global optimum
- ◇ reservoir (good spots)
- ✕ abandoned basin
- re-ignite re-seed
- repelled re-seed

① Informed restart — how it works

Re-seed from what the search already found, not blindly

REPELLED

$$x' \sim \text{Unif} \text{ s.t. } \|x' - c\| > 0.1 s$$



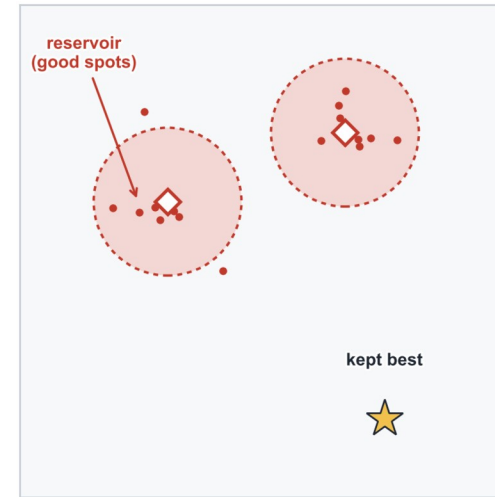
uniform, but repelled from remembered basins



no improvement 300 evals
&
 $f_{\text{best}}/|f_{\text{init}}| > 1e-8$

RE-IGNITE

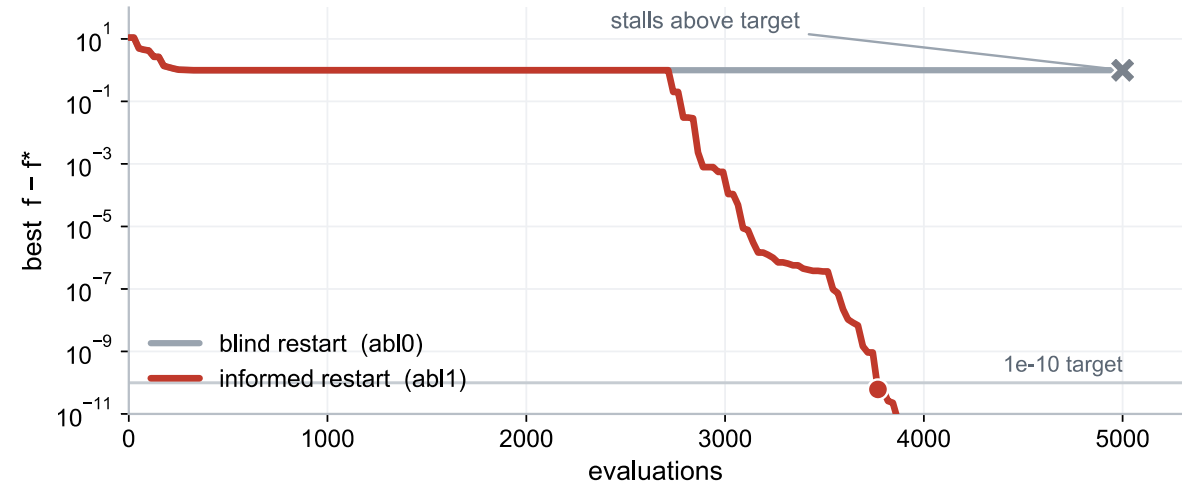
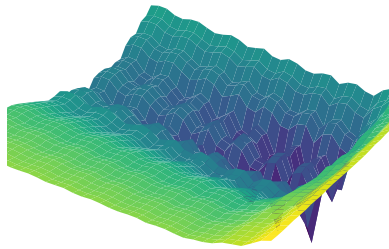
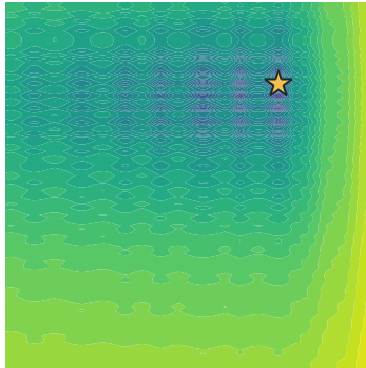
$$x' \sim \mathcal{N}(x_{\text{archive}}, (0.05 s)^2)$$



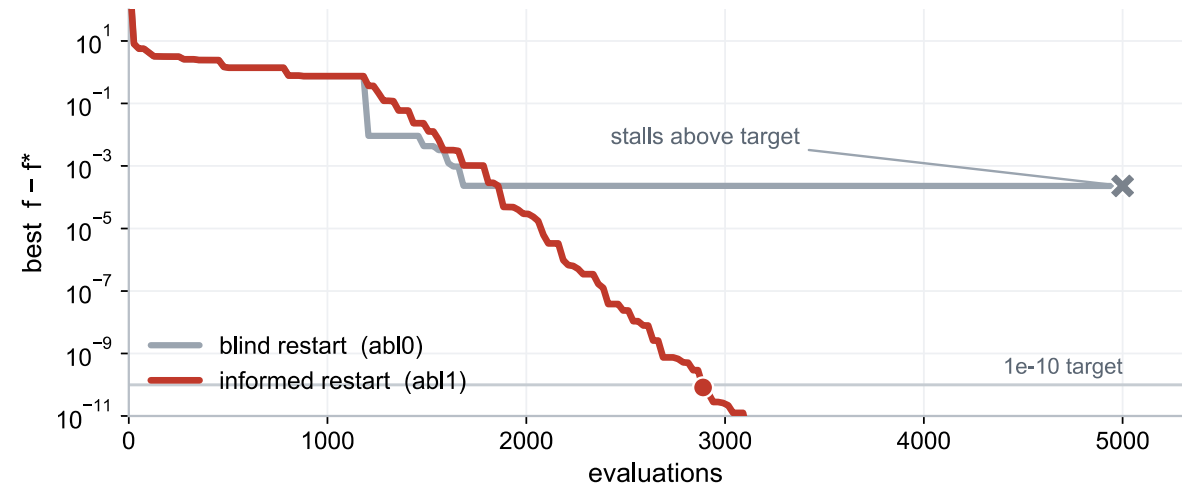
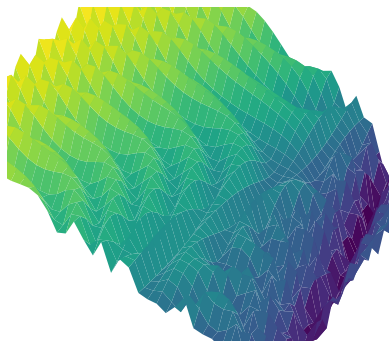
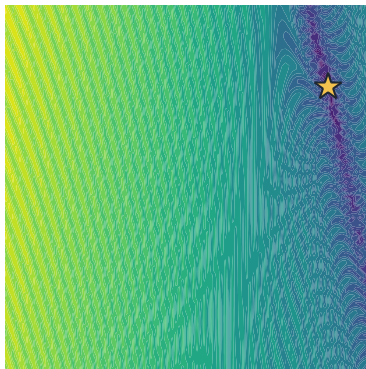
re-light around archived elites — good spots found

① Informed restart — result

F04-BucheRastrigin / seed 200



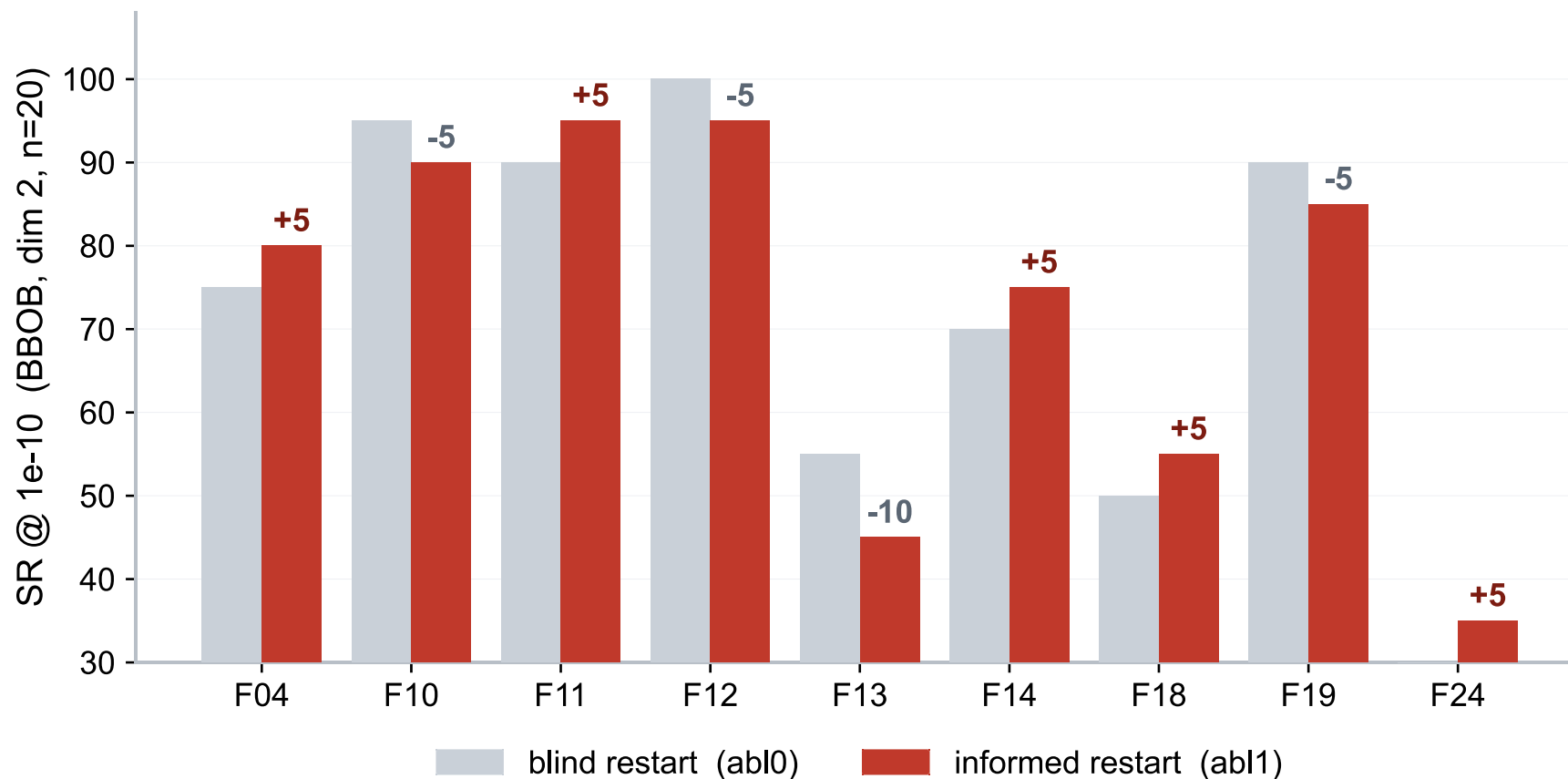
F18-SchafferF7ill / seed 1100



① Informed restart — per-function SR

Net ± 0 on BBOB-24 — the payoff is multimodal escape

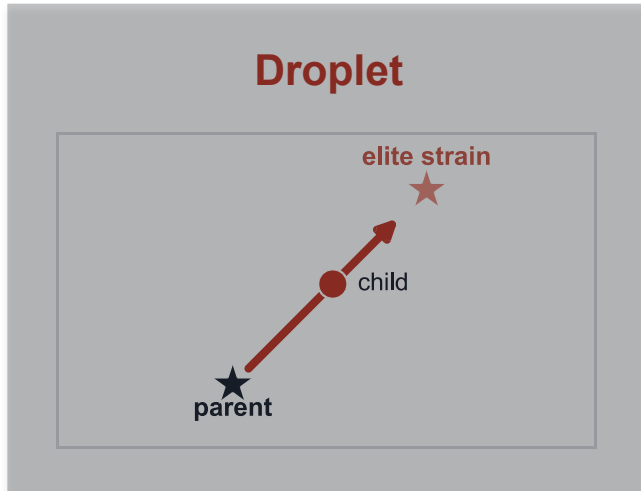
change-ablation / BBOB-24 / dim 2 / n = 20



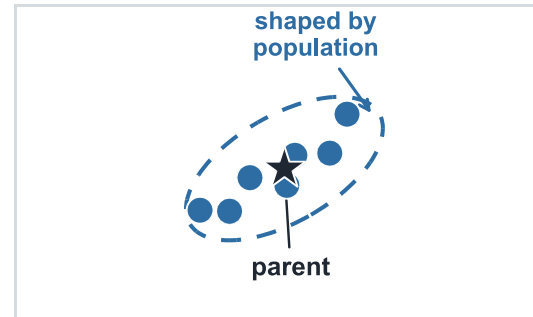
② Adaptive anisotropy floor

Transmission channels

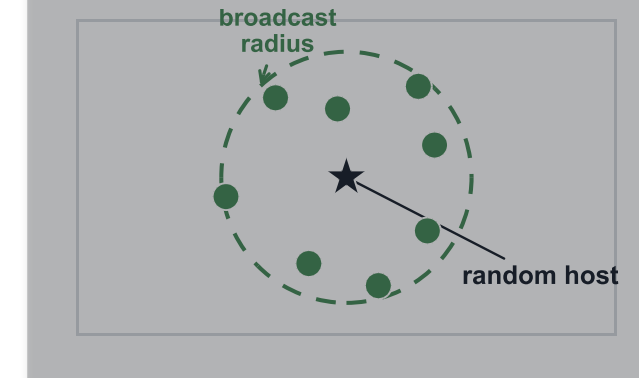
three parallel reproduction routes



Close-contact

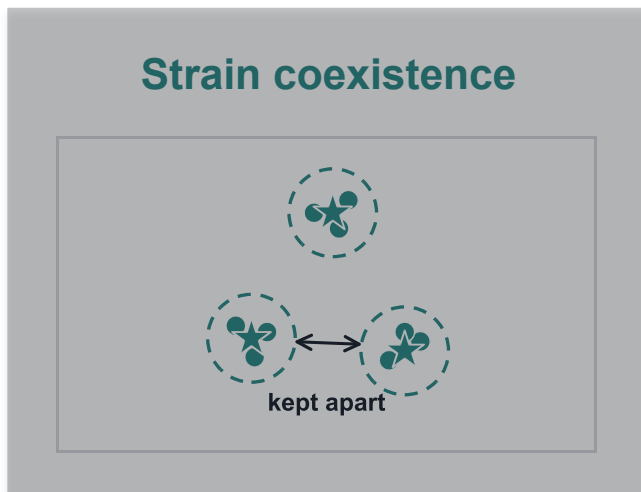


Airborne



Population strategies

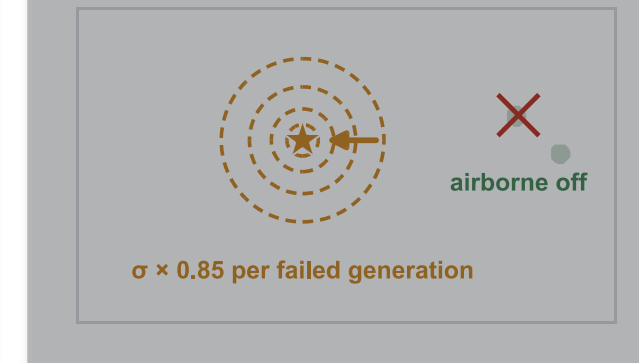
how the swarm is steered



Restart (spillover)



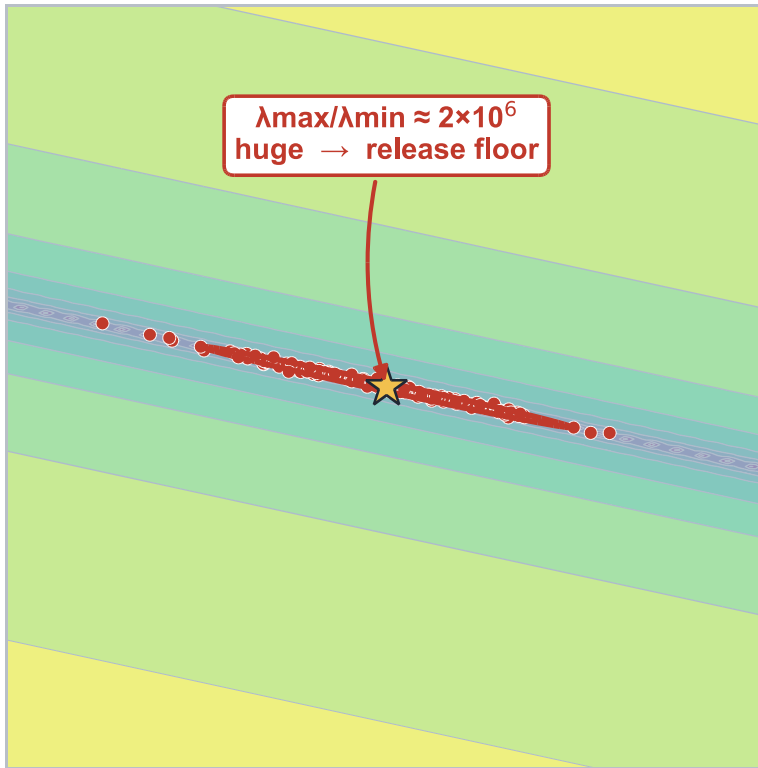
Drilling



② Adaptive anisotropy floor

Close-contact noise adapts to the landscape shape

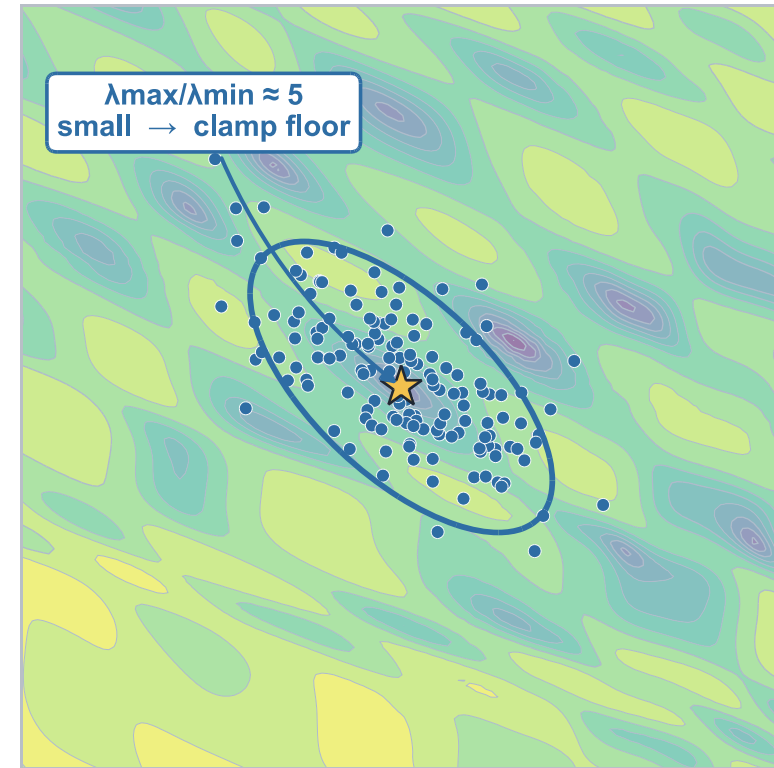
Ill-conditioned valley (F10)



release anisotropy $\rightarrow$ align to the valley / Gaussian axis ratio 45 : 1

★ parent (host) ● offspring ○ Gaussian 2σ

Rugged multimodal (F15)



clamp anisotropy $\rightarrow$ stay round & cautious / Gaussian axis ratio 2 : 1

★ parent (host) ● offspring ○ Gaussian 2σ

② Adaptive anisotropy floor — the math

Only the floor φ is new — the pipeline is unchanged

Close-contact channel — same flow as before

EIGEN-DECOMPOSE

$$C = V \Lambda V^T$$

population spread $\rightarrow$ axes V , variances λ

FLOOR THE AXES

$$\hat{\lambda}_i = \max(\tilde{\lambda}_i, \varphi)$$

lift any collapsed axis up to the floor φ

SAMPLE CHILD

$$x' = x_p + \sigma_i V \sqrt{\hat{\Lambda}} z$$

draw from the floored ellipse (step σ_i , noise z)

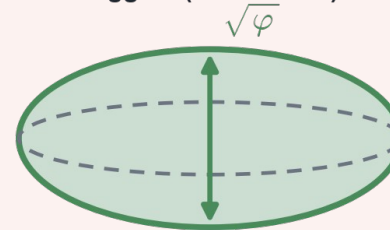
The floor φ is now adaptive

$$r = \lambda_{\max} / \lambda_{\min} \rightarrow \rho = \text{EMA}(\log_{10} r)$$

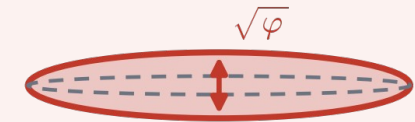
EMA = exponential moving average — smooths out transient blips

$$\varphi(\rho) = \varphi_{hi} (= 10^{-2}) \rightarrow \varphi_{lo} (= 10^{-3})$$

rugged (multimodal)

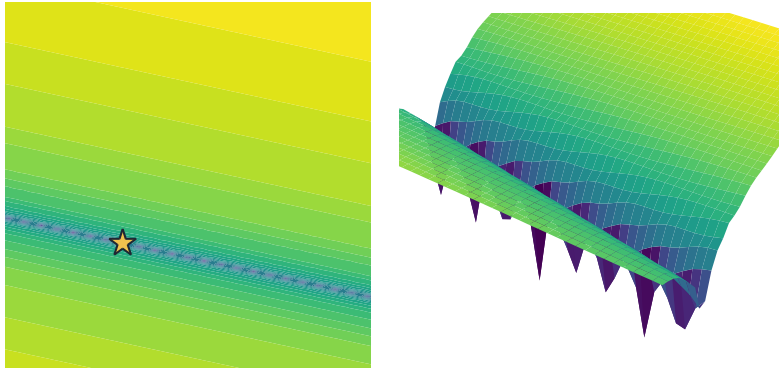


ill-conditioned

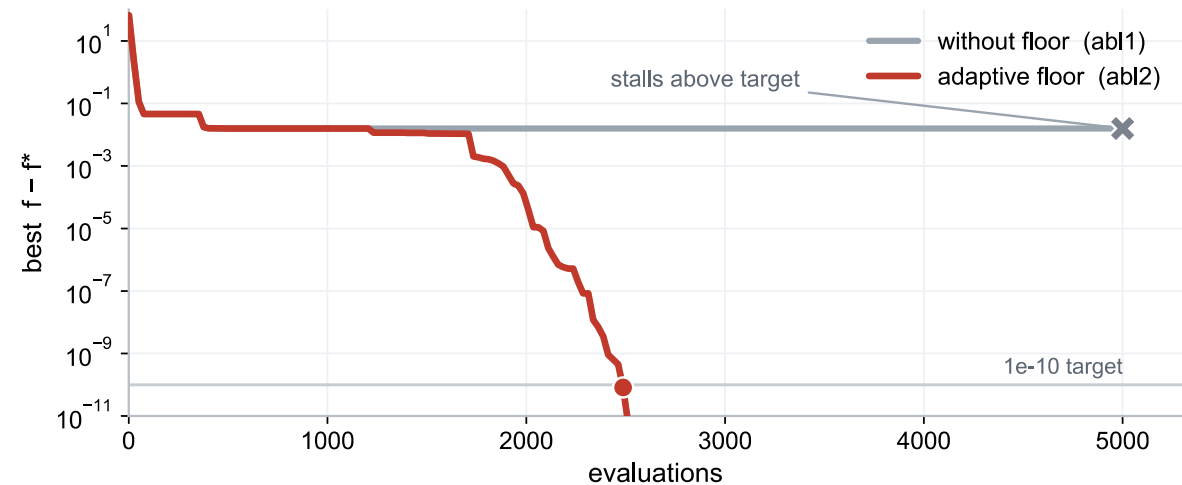
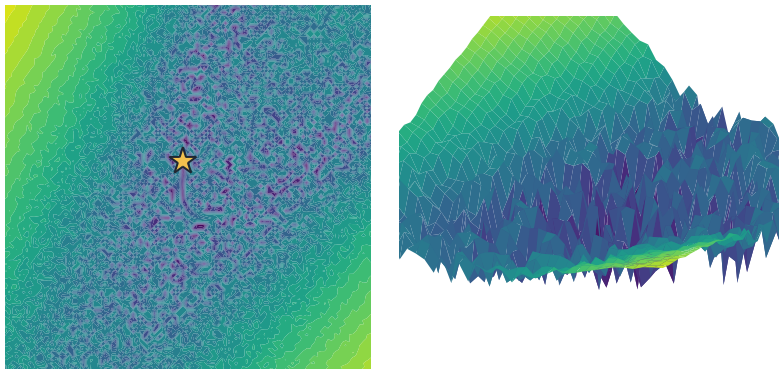


② Adaptive anisotropy floor — result

F10-EllipsoidalRot / seed 300



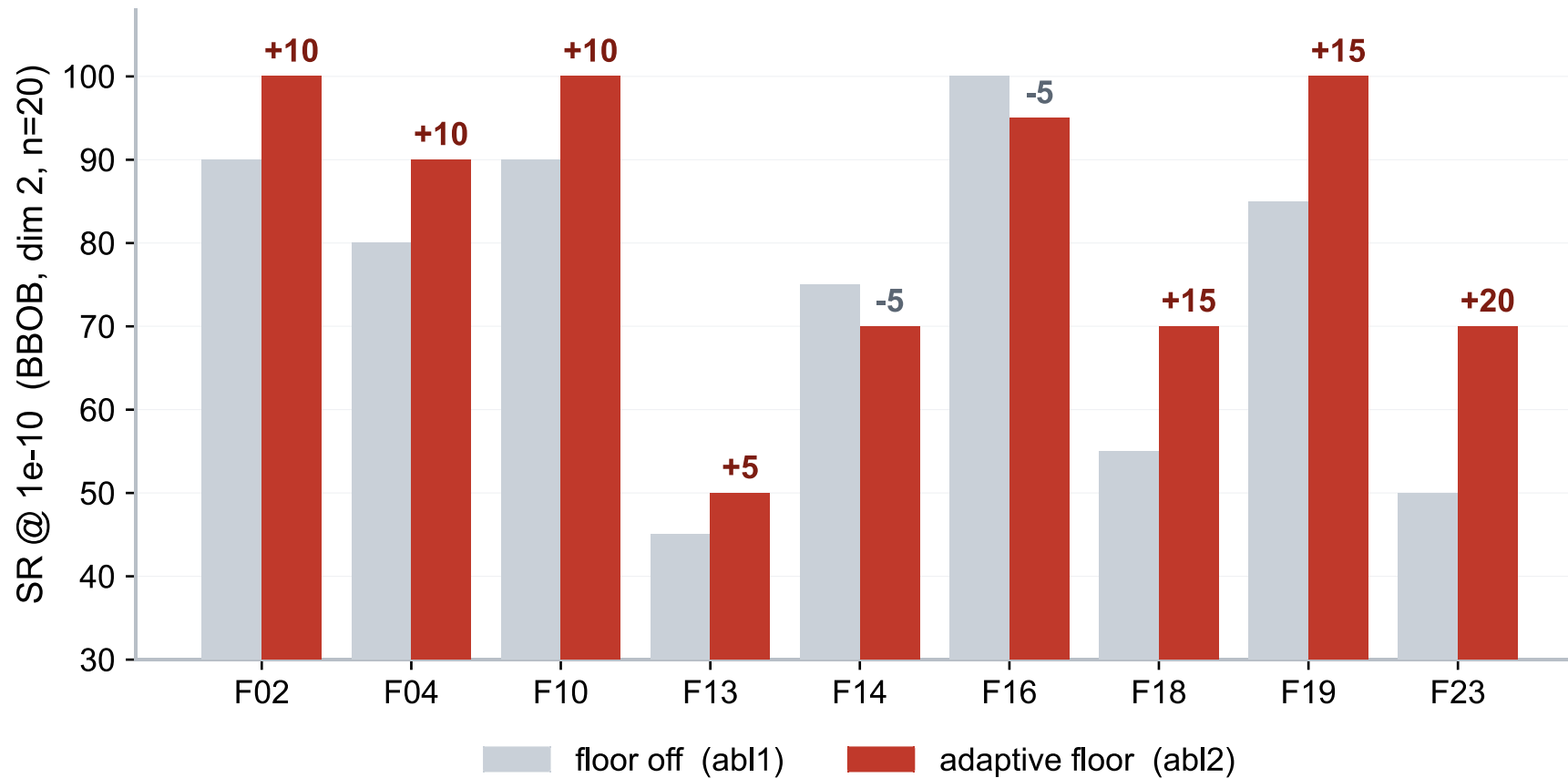
F19-GriewankRosenbrock / seed 1400



② Adaptive anisotropy floor — per-function SR

Net +3.1 pt — the largest single step

change-ablation / BBOB-24 / dim 2 / n = 20 / functions whose SR@1e-10 moved

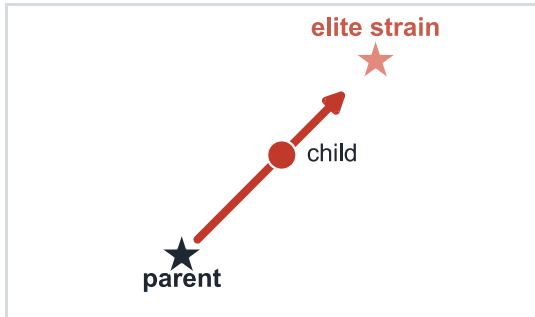


③ Channel router

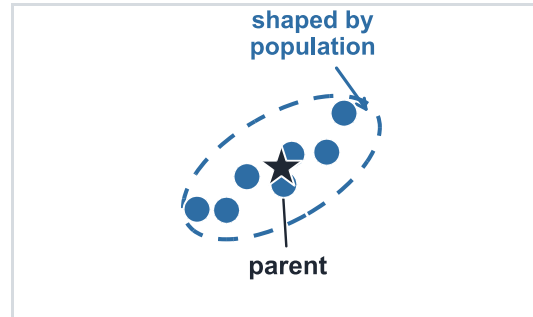
Transmission channels

three parallel reproduction routes

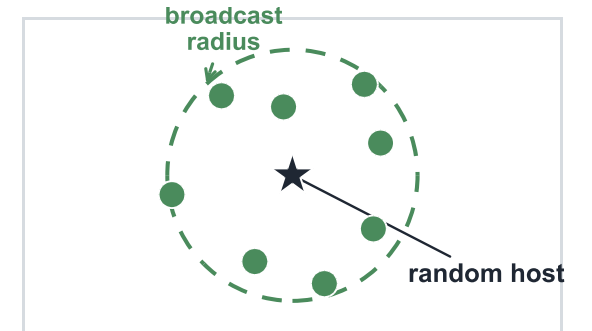
Droplet



Close-contact



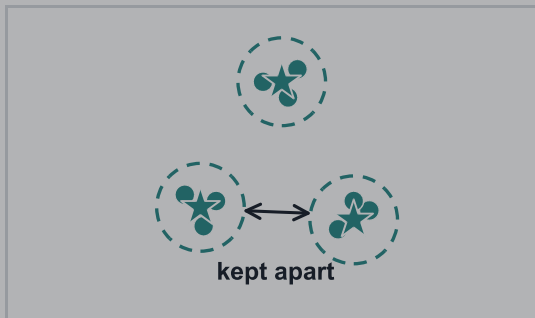
Airborne



Population strategies

how the swarm is steered

Strain coexistence



Restart (spillover)



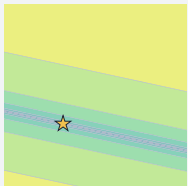
Drilling



③ Channel router

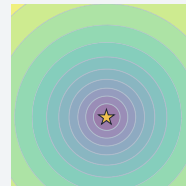
Each run locks one route, read off its covariance shape

DROPLET



$$\text{cond} = \log_{10} \frac{\lambda_{\max}}{\lambda_{\min}} > 3$$

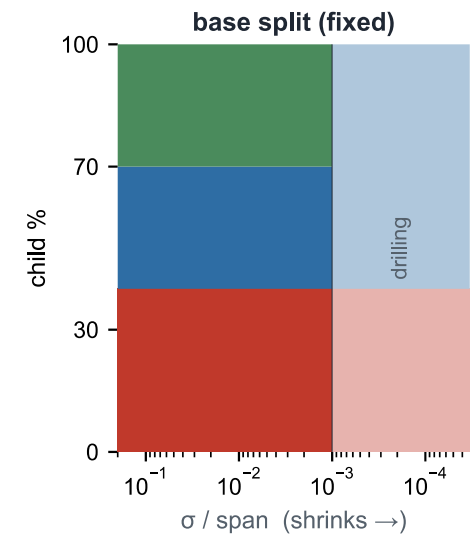
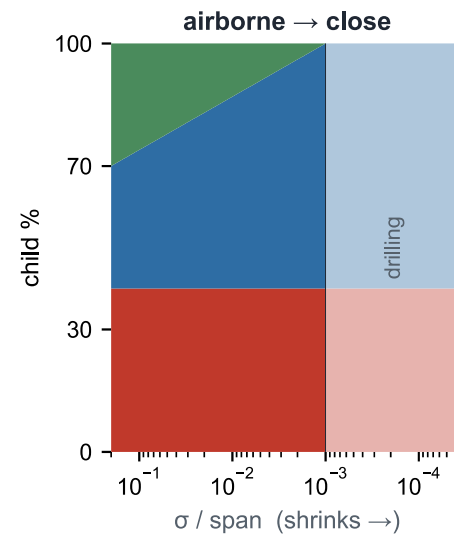
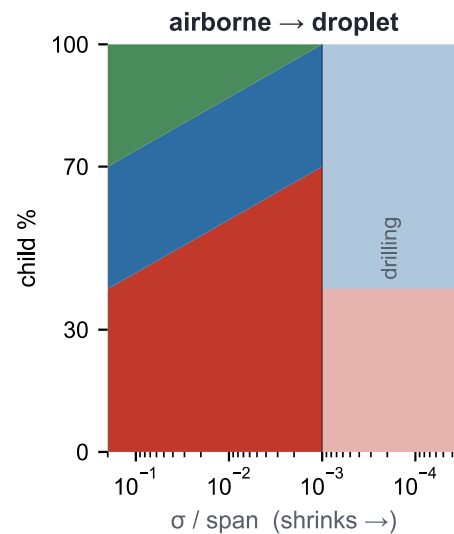
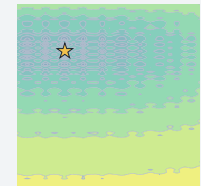
CLOSE



$$\text{algA} = \left\langle \max_j |V_{ij}| \right\rangle_i \geq .965$$

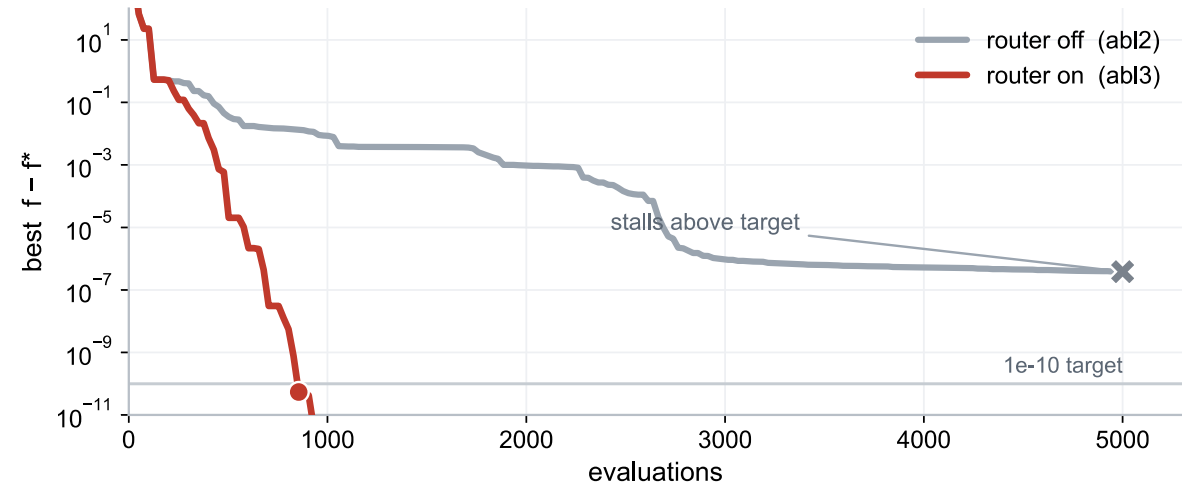
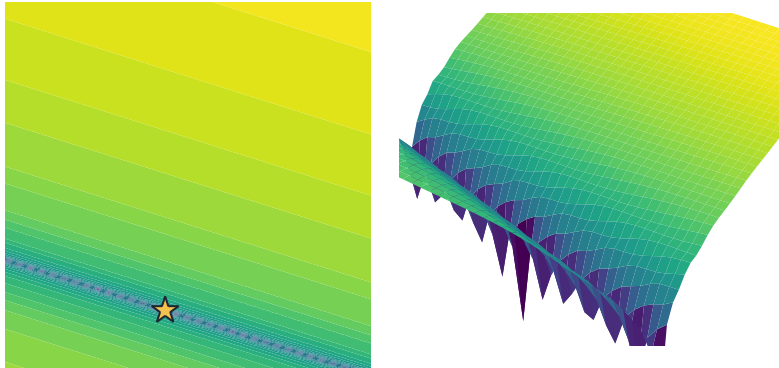
$$\text{mgap} = \max_i \text{gap}_i / \text{range}_i \geq .36$$

KEEP-AIR

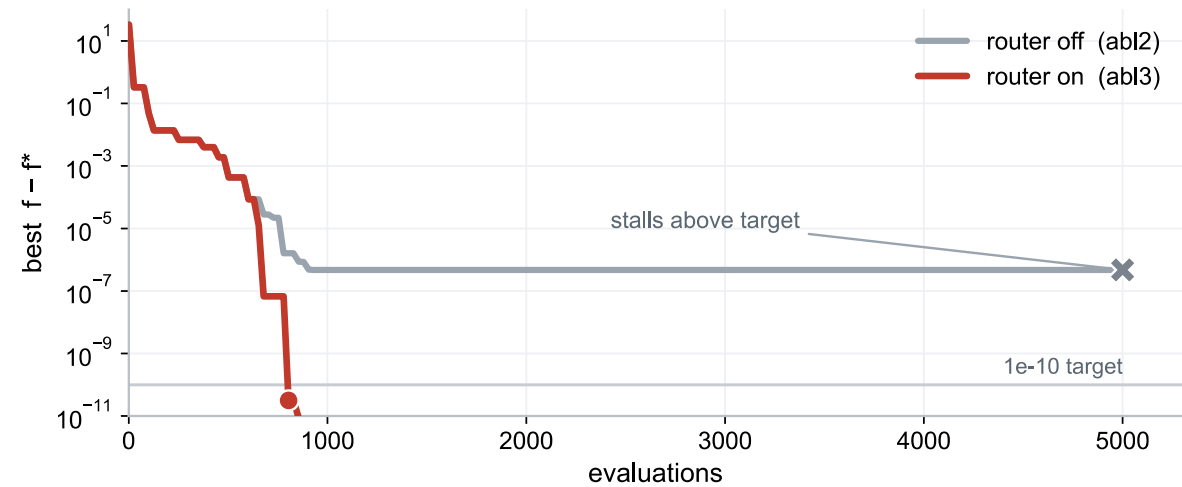
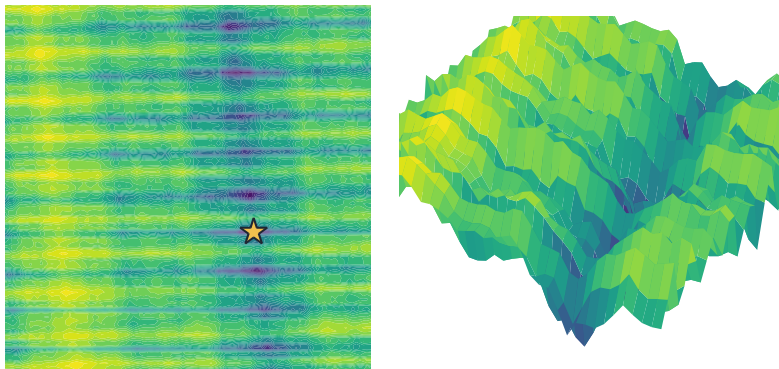


③ Channel router — result

F11-Discus / seed 600



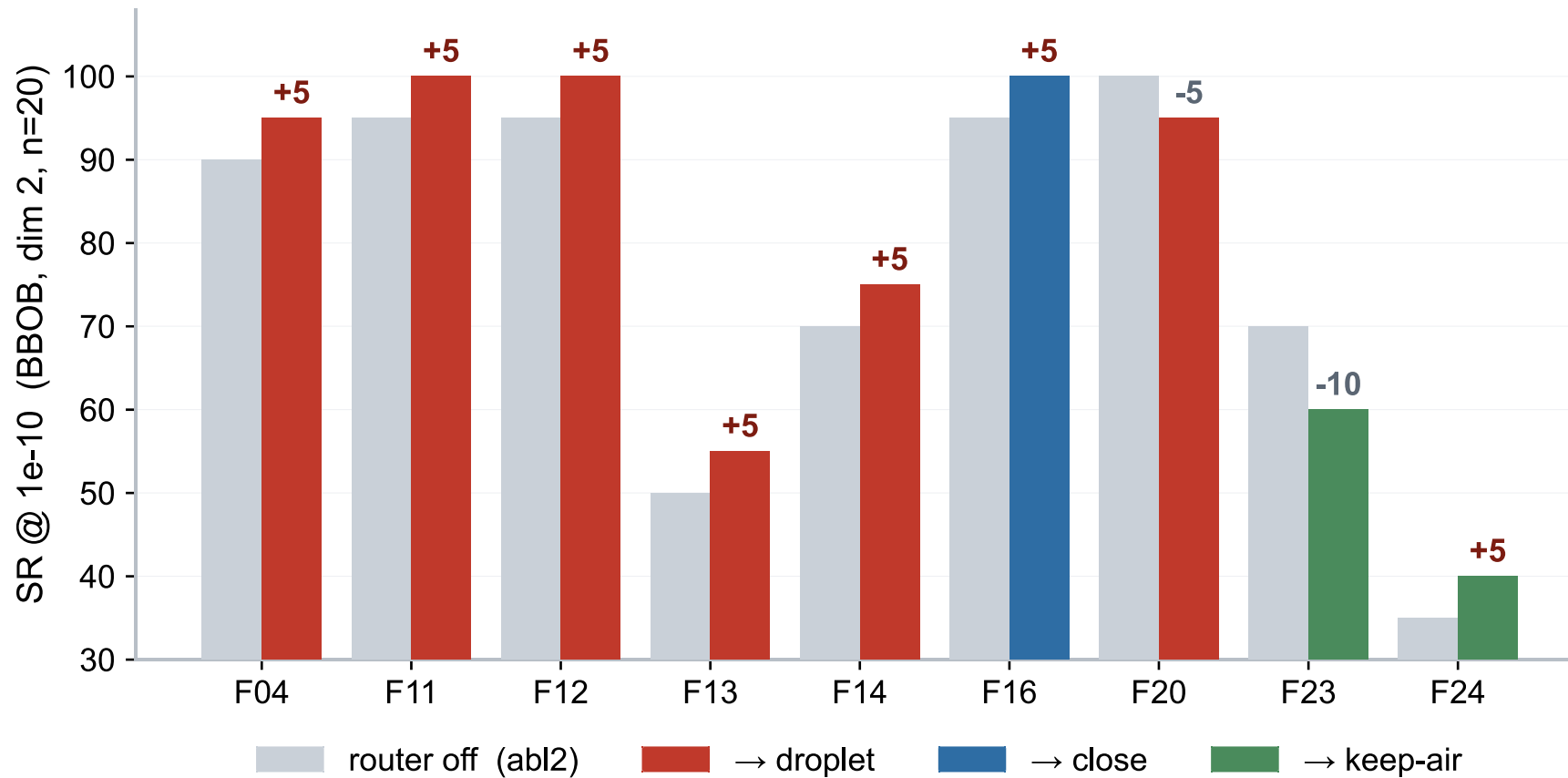
F16-Weierstrass / seed 1200



③ Channel router — per-function SR

Lifts exactly what it routes — net +0.8 pt

change-ablation / BBOB-24 / dim 2 / n = 20 / functions whose SR@1e-10 moved

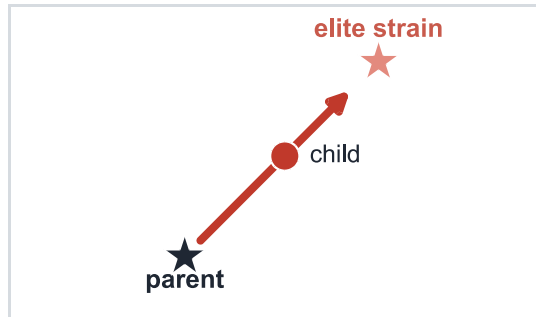


④ Route-gated best2

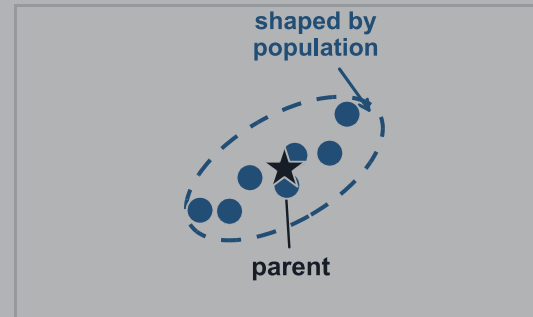
Transmission channels

three parallel reproduction routes

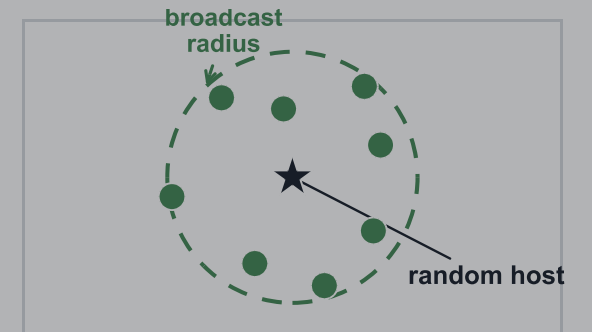
Droplet



Close-contact



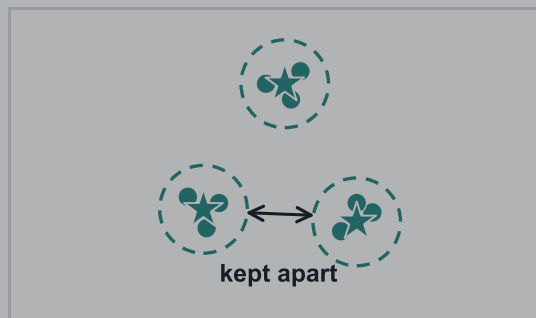
Airborne



Population strategies

how the swarm is steered

Strain coexistence



Restart (spillover)

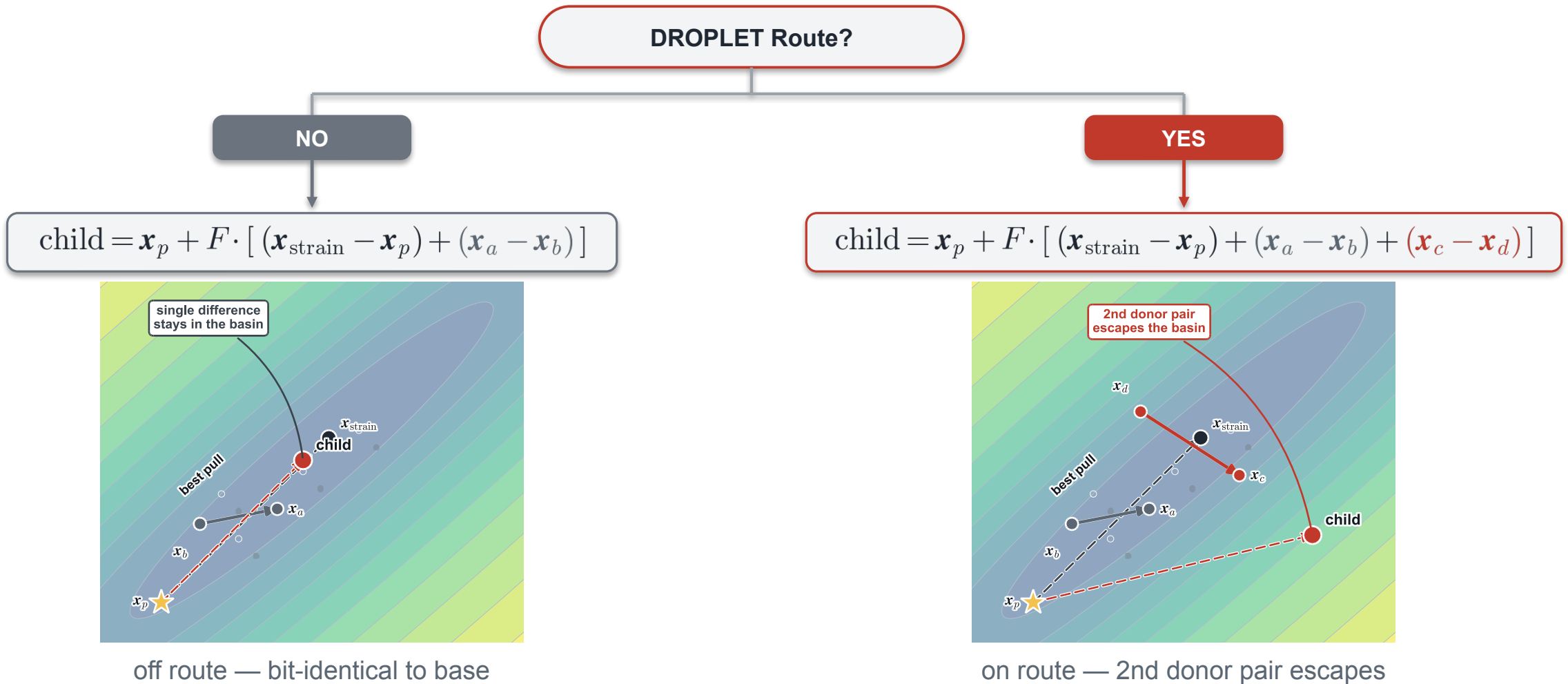


Drilling



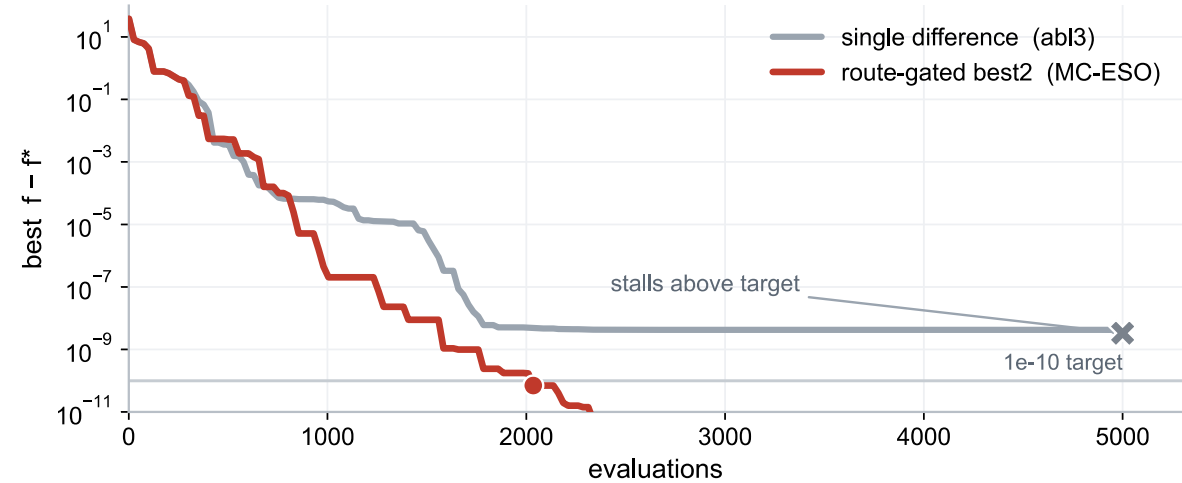
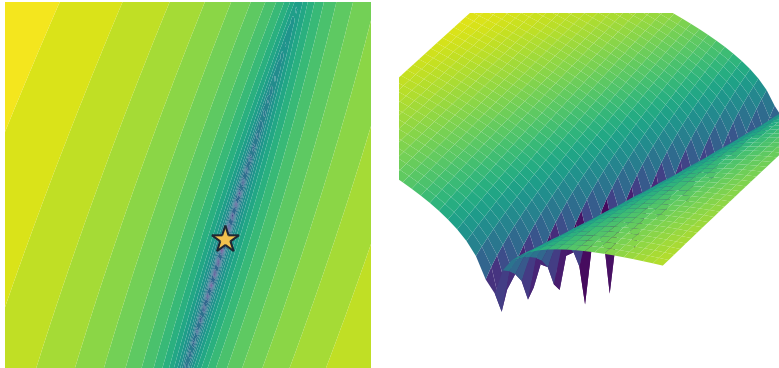
④ Route-gated best2

A 2nd difference vector — only on droplet-routed runs

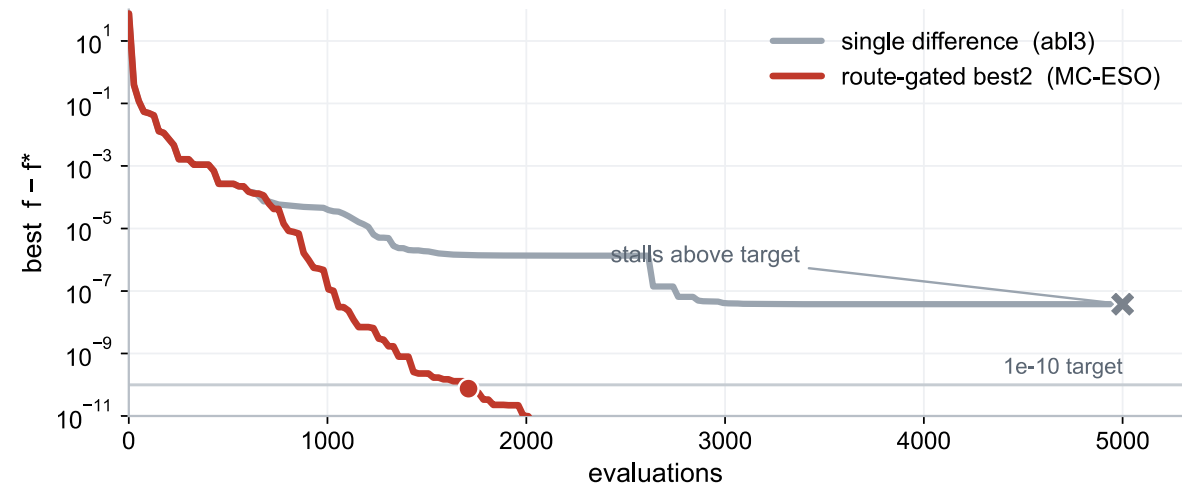
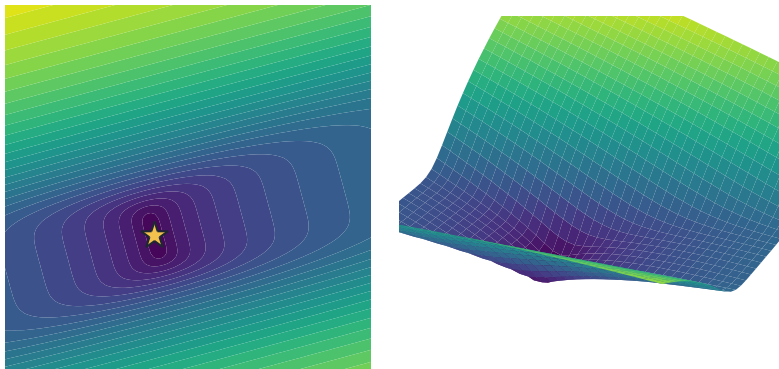


④ Route-gated best2 — result

F13-SharpRidge / seed 300



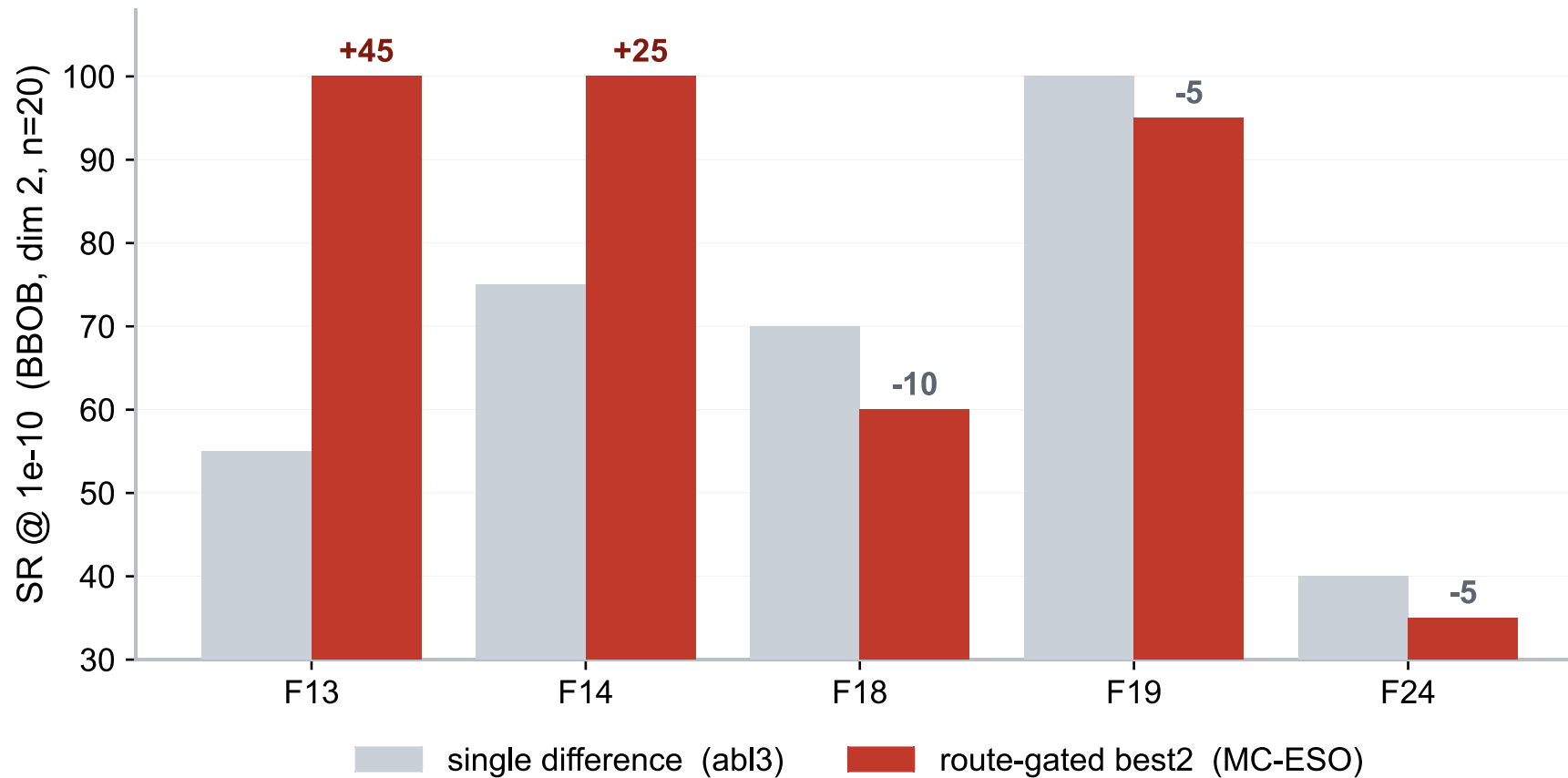
F14-DiffPowers / seed 500



④ Route-gated best2 — per-function SR

F13/F14 jump to 100% — net +2.1 pt

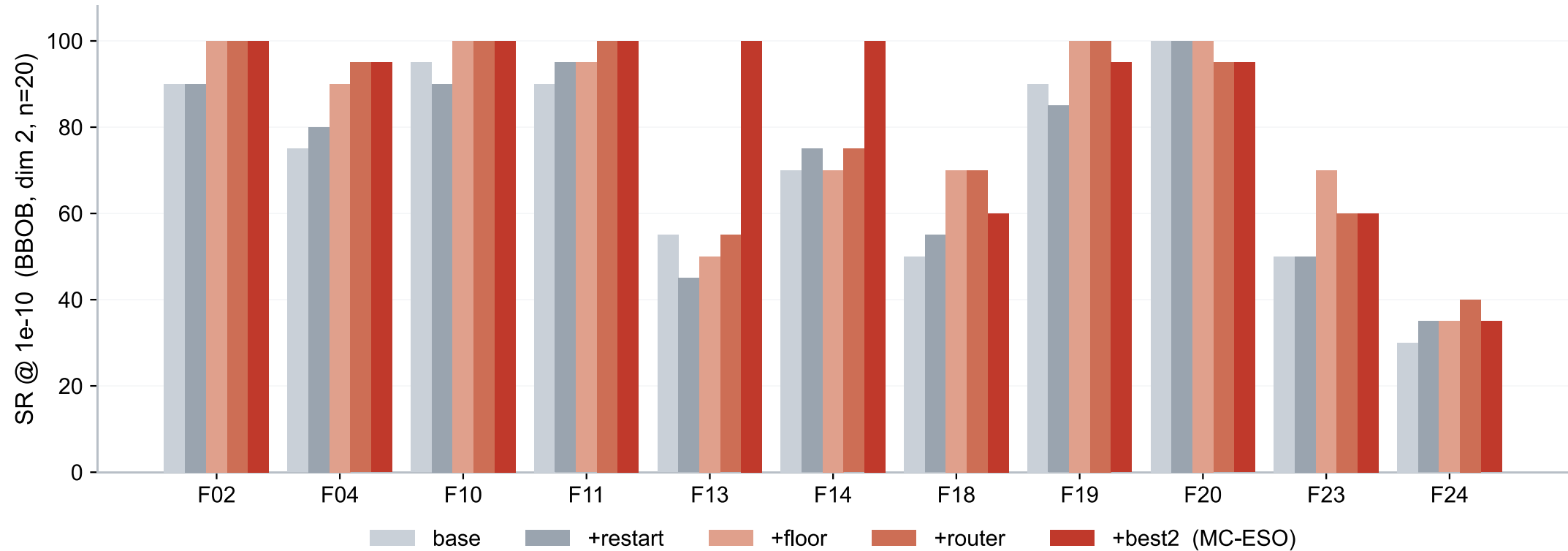
change-ablation / BBOB-24 / dim 2 / n = 20 / functions whose SR@1e-10 moved



The four changes together

SR@1e-10 by function, base → MC-ESO

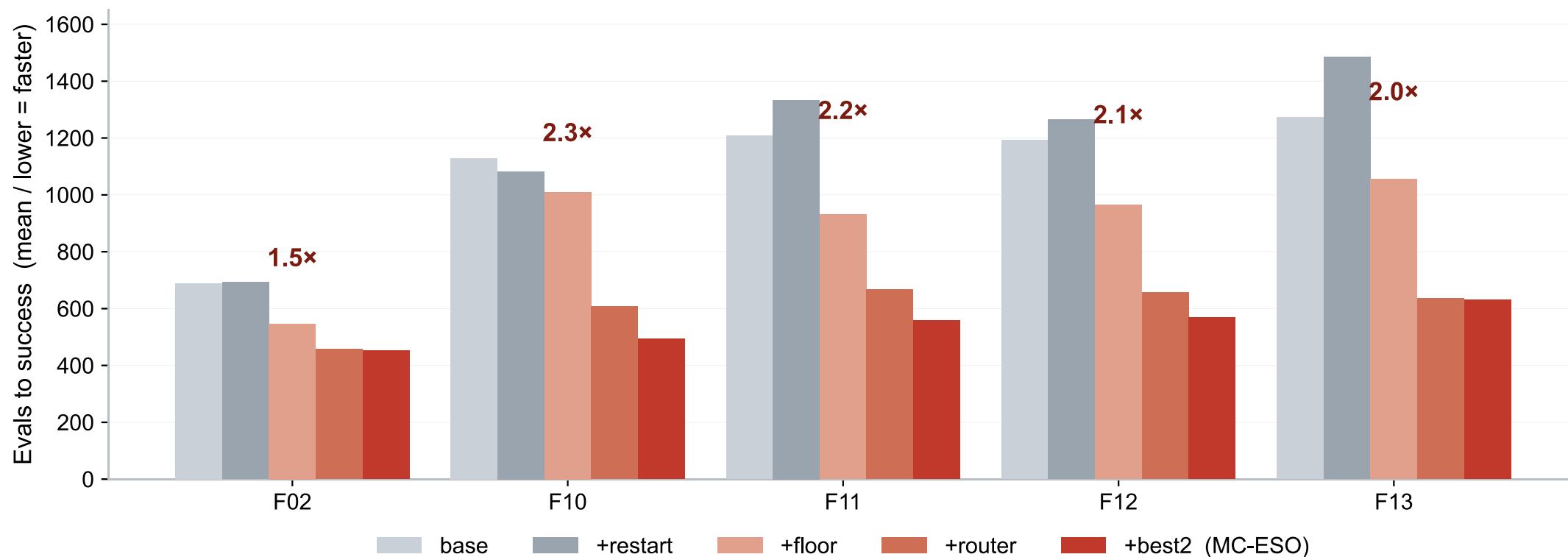
change-ablation / BBOB-24 / dim 2 / n = 20 / base → +restart → +floor → +router → +best2



Ill-conditioned valleys converge $\sim 2\times$ faster

Same SR, about half the evaluations

evals-to-success / same ablation ladder / BBOB-24 / dim 2 / n = 20



4

Comparison

MC-ESO vs existing methods

The comparison set

Five baselines — colours match the curves that follow



IPOP-CMA-ES [1]

CMA-ES with restarts + growing population

weak: multimodal



PSO [2]

Inertia-weighted particle swarm

weak: ill-conditioned



DE [3]

DE/rand/1/bin — the droplet channel's ancestor

weak: multimodal



SaVOA [4]

Self-adaptive virus optimization

weak: ill-conditioned



NM-Restart [5]

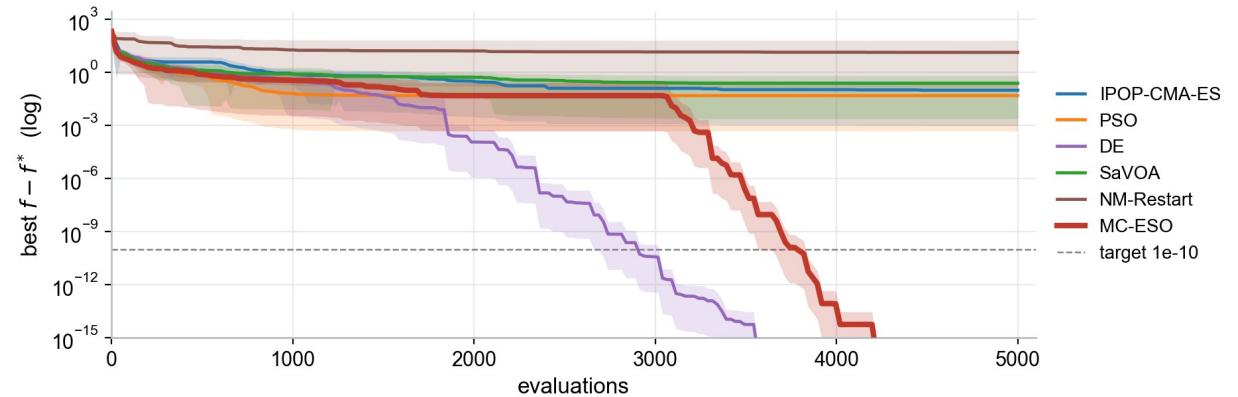
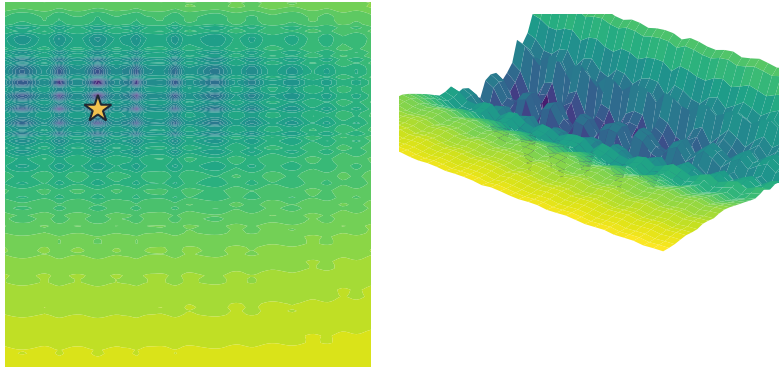
Restarted Nelder–Mead simplex

strong: low-dim

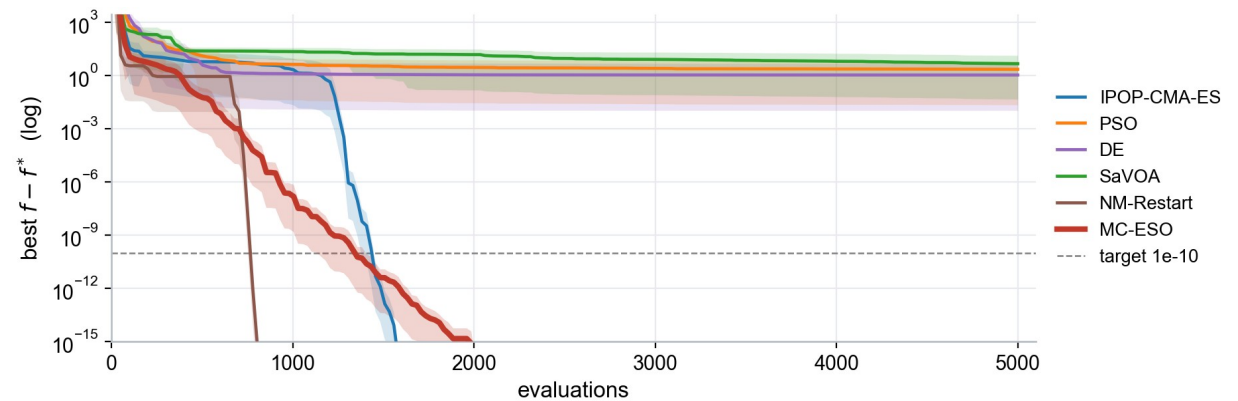
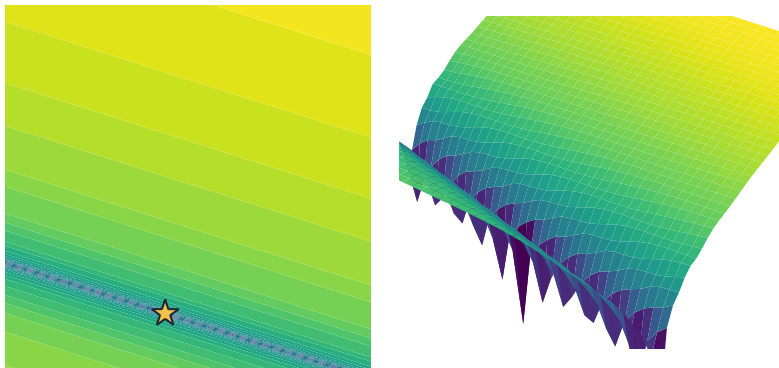
Others (CMA-ES / BIPOP / L-SHADE / NCDE) → appendix

Where baselines struggle — multimodal & ill-conditioned

F03 / Rastrigin (separable) — multimodal

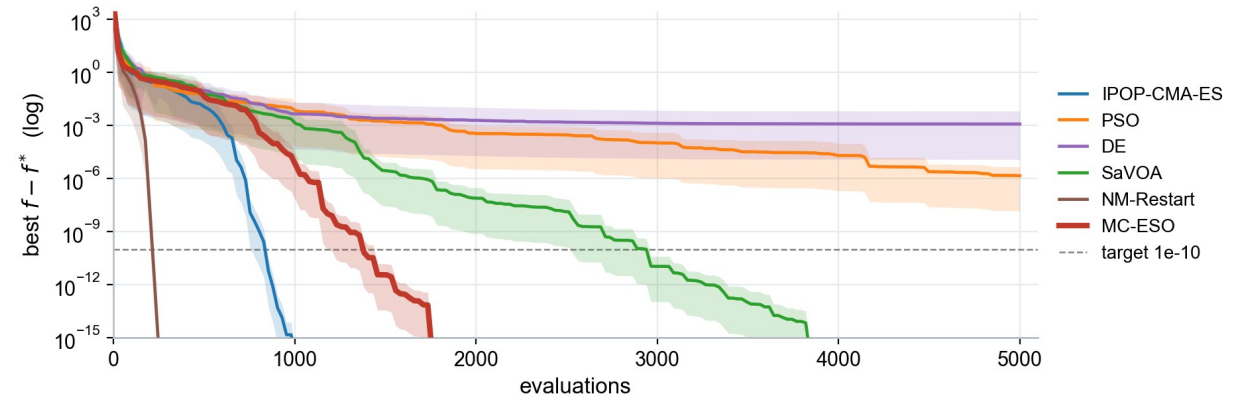
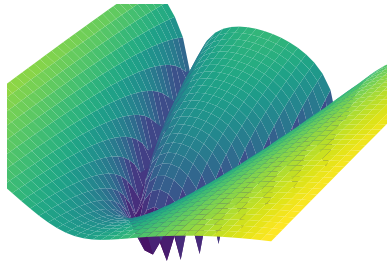
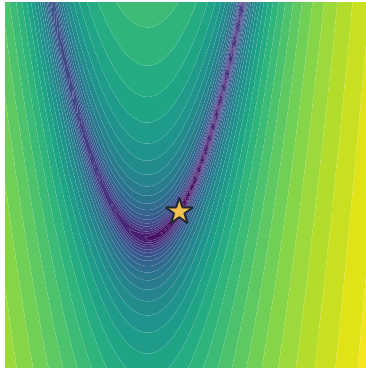


F11 / Discus — ill-conditioned

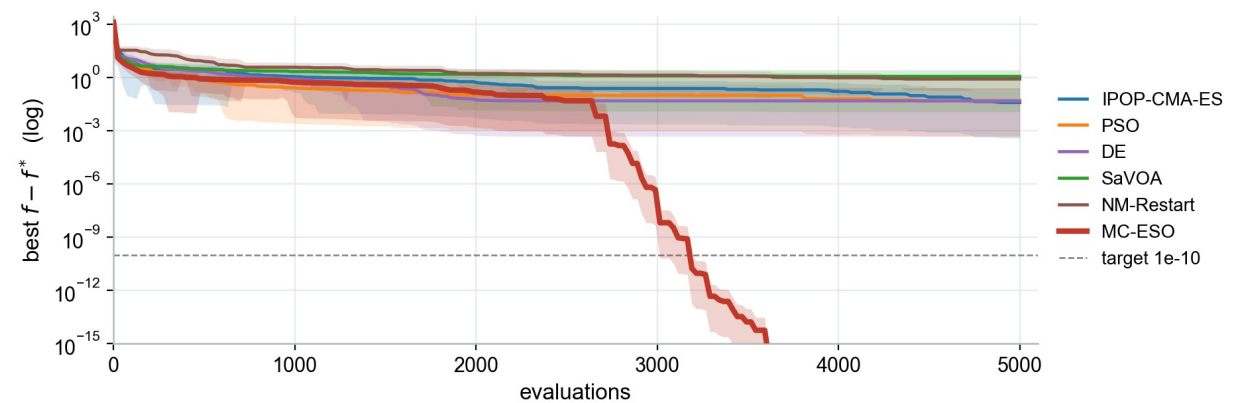
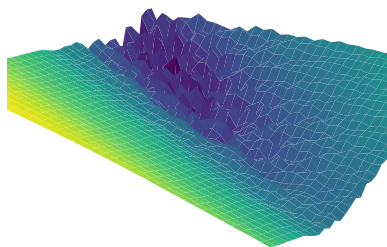
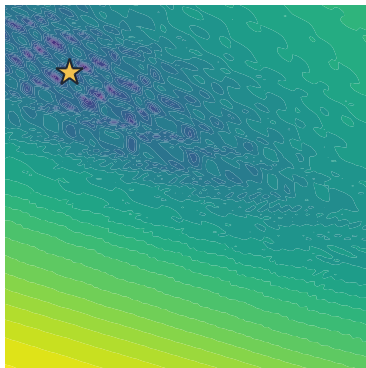


Where baselines struggle — bent valley & hardest case

F08 / Rosenbrock — bent valley



F15 / Rastrigin (rotated) — hardest



Standing today — first at deep precision

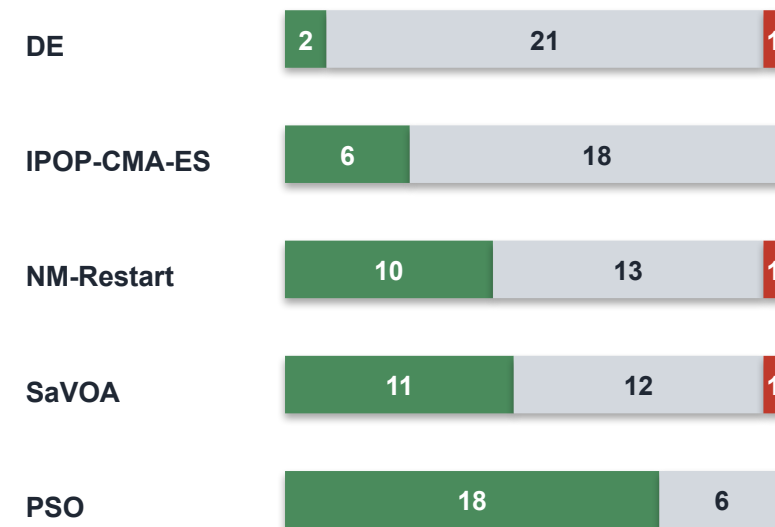
SR@1e-10 92.9% — DE overtaken

each cell: SR@1e-10 (left) / evals-to-success (right) / BBOB-24 / dim 2 / n = 20

	Separable F01–05	Moderate F06–09	Ill-cond. F10–14	Multimodal F15–19	Weak-struct. F20–24	ALL F01–24
MC-ESO	99 / 851	100 / 496	100 / 528	89 / 934	78 / 1080	93 / 790
DE	98 / 1043	91 / 924	98 / 1025	94 / 1954	66 / 1741	89 / 1338
IPOP-CMA-ES	77 / 947	100 / 312	99 / 381	90 / 1556	54 / 1466	83 / 936
NM-Restart	69 / 1147	76 / 682	100 / 145	15 / 2139	67 / 1120	65 / 956
SaVOA	86 / 1018	100 / 767	18 / 2716	49 / 1731	55 / 1839	60 / 1650
PSO	89 / 1369	52 / 1419	7 / 1806	50 / 2200	43 / 1751	48 / 1712

Wilcoxon — vs each baseline

win tie loss



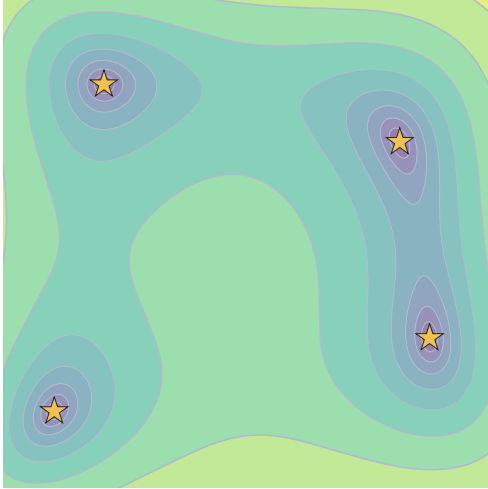
5

Multimodality

The deferred axis — partial progress

Multimodality — SR hides the gap

SR is 100%, yet few optima are found (low PR)

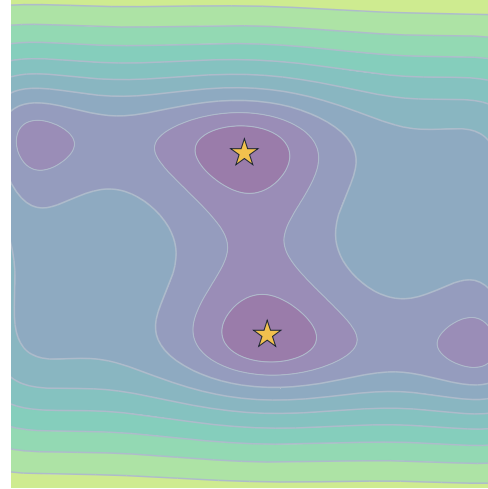


C01 Himmelblau

K = 4 optima

SR@1e-10 **100%**

PR@1e-4 **0.28**

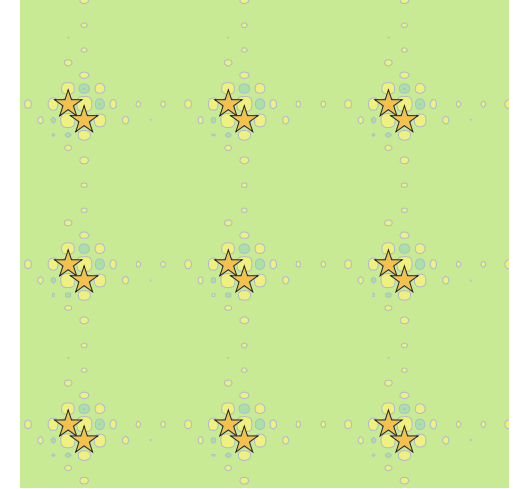


C02 Six-hump

K = 2 optima

SR@1e-10 **100%**

PR@1e-4 **0.60**



C03 Shubert

K = 18 optima

SR@1e-10 **100%**

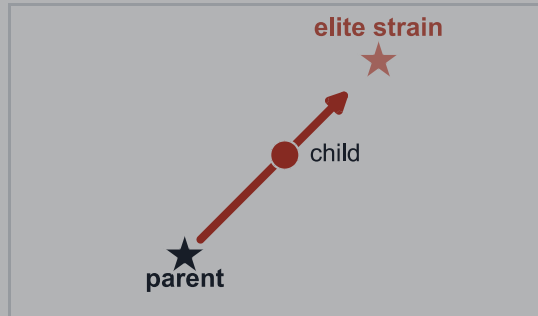
PR@1e-4 **0.06**

Sequential niching

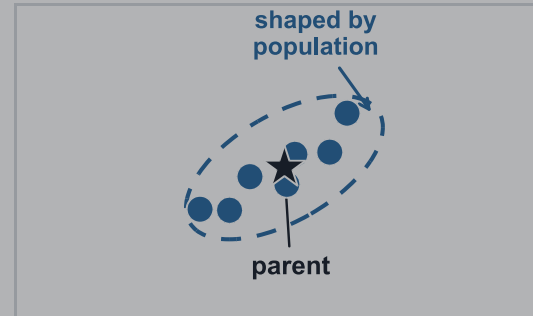
Transmission channels

three parallel reproduction routes

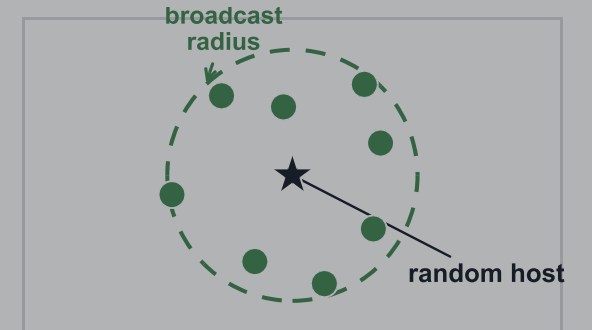
Droplet



Close-contact



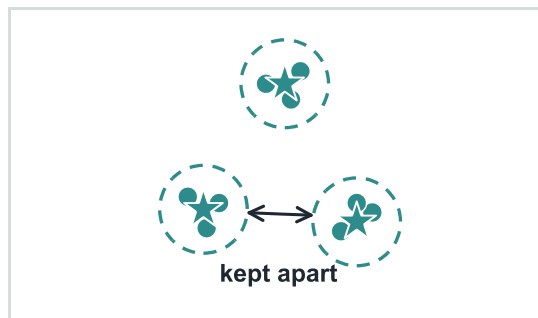
Airborne



Population strategies

how the swarm is steered

Strain coexistence



Restart (spillover)

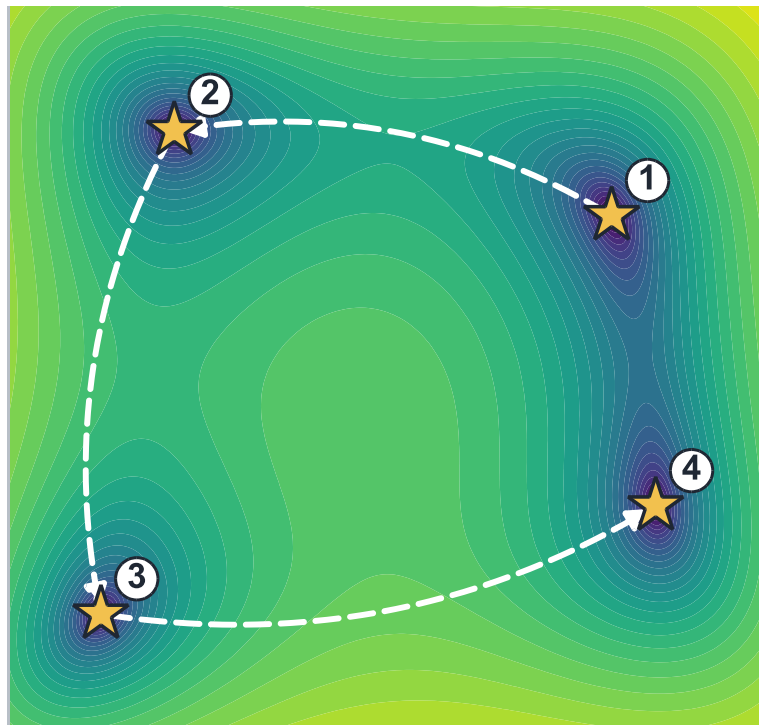


Drilling



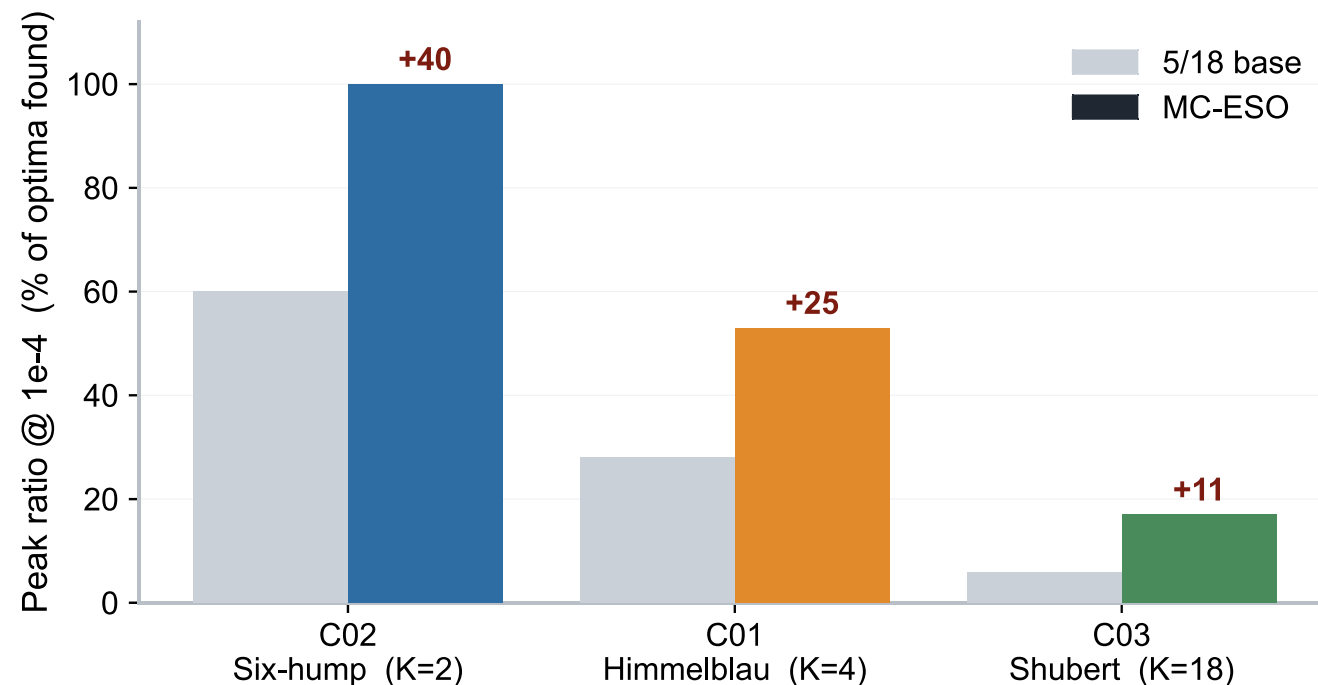
Sequential niching — more optima, SR untouched

One run captures optima one at a time — SR@1e-10 stays 100%



① **drill a basin to 1e-10** — like base

② **σ -exhausted** → **repelled restart** to the next optimum



6

Wrap-up

Summary and what remains

Summary

The whole cycle in three lines

Idea

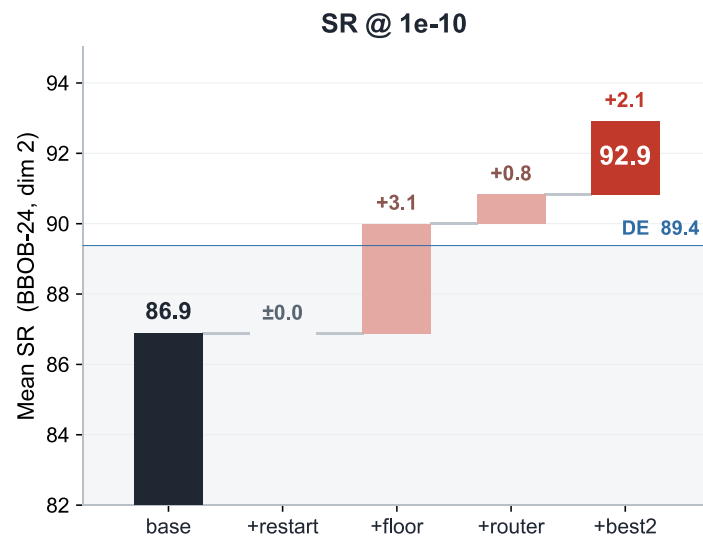
- › Uniform channel mix →
- › landscape-aware routing

Mechanism

- › informed restart
- › adaptive floor
- › channel router
- › route-gated best2

Result

- › SR@1e-10 **86.9 → 92.9%**
- › top of 10 methods
- › valleys 2× faster
- › multi-solution ↑, SR kept



	Separable F01–05		Moderate F06–09		Ill-cond. F10–14		Multimodal F15–19		Weak-struct. F20–24		ALL F01–24	
MC-ESO	99	851	100	496	100	528	89	934	78	1080	93	790
DE	98	1043	91	924	98	1025	94	1954	66	1741	89	1338
IPOP-CMA-ES	77	947	100	312	99	381	90	1556	54	1466	83	936
NM-Restart	69	1147	76	682	100	145	15	2139	67	1120	65	956
SaVOA	86	1018	100	767	18	2716	49	1731	55	1839	60	1650
PSO	89	1369	52	1419	7	1806	50	2200	43	1751	48	1712

Remaining work

Three directions

Generalize the constants

hard-coded thresholds — one set per improvement

1 restart	300 evals / $1e-8$
2 floor	ϕ $1e-2 \dots 1e-3$
3 router	cond > 3 / algA $\geq .965$ / mgap $\geq .36$



derive from problem scale

thresholds may be BBOB-tuned — make them scale-derived

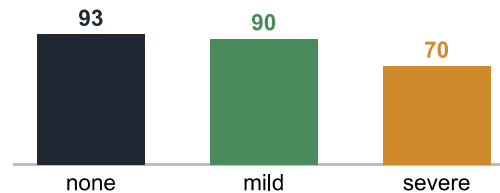
Robustness

MULTI-SOLUTION



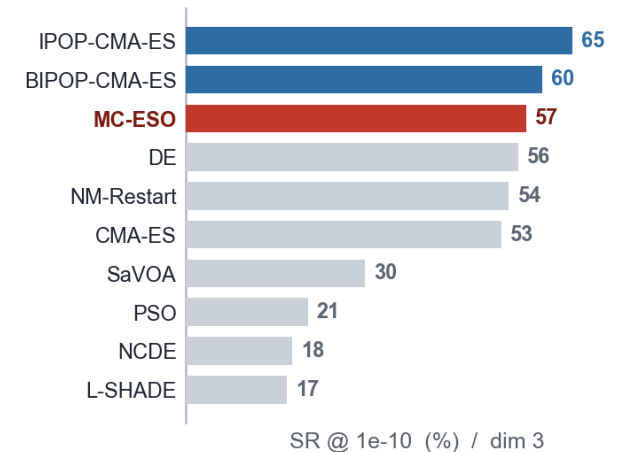
finds 1 of many

UNDER NOISE — SR@ $1e-10$



partial multi-solution / robust to mild noise, severe open

Higher dimensions



at 3D the CMA-ES restart family overtakes MC-ESO at $1e-10$

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- [3] R. Storn, K. Price. "Differential evolution." J. Global Optimization, 1997.
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- [6] N. Hansen, A. Ostermeier. "Completely derandomized self-adaptation in evolution strategies." Evolutionary Computation, 2001.
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- [9] B.-Y. Qu, P. N. Suganthan, J.-J. Liang. "DE with neighborhood mutation for multimodal optimization." IEEE TEVC, 2012.
- [10] N. Hansen, A. Auger, R. Ros, O. Mersmann, T. Tušar, D. Brockhoff. "COCO: comparing continuous optimizers in a black-box setting." Optim. Methods & Software, 2021.

Appendix — the other baselines

In the full comparison; omitted from charts for legibility

CMA-ES [6]

Plain covariance-matrix ES with multistart — the IPOP base without population doubling.

BIPOP-CMA-ES [7]

CMA-ES restarts alternating large / small population regimes to keep the budgets balanced.

L-SHADE [8]

Success-history adaptive DE with linear population reduction (CEC2014 winner).

NCDE [9]

Neighborhood-based crowding DE — a niching specialist, used only for the multi-optima (PR / MMOsr) comparison.

Appendix — landscape-signal measurements

All 35 functions in BBOB category order — the diagnostic behind the channel router

cond = log10 eigenvalue ratio, **PR** = participation ratio [1,2], **algA** = axis alignment [0,1], **offd** = corr. off-diag RMS [0,1], **divs** = variance/span, **kurt** = excess kurtosis, **mgap** = max normalized gap, **nelt** = mean niche count, **spil** = spillover count **values** = median over exploration generations ($\sigma > \text{drilling}$)

highlighted cells = router triggers: **cond** $\geq 3.0 \rightarrow$ **droplet** (checked first) **algA** $> 0.965 \wedge$ **mgap** $> 0.36 \rightarrow$ **close** else $\rightarrow$ **keep-air** (= base)

base MC-ESO (unmodified) / dim 2 / n = 8

function	route	cond	PR	algA	offd	divs	kurt	mgap	nelt	spil
Separable — F01–05										
F01-Sphere	-	0.38	1.71	0.917	0.268	0.002	-0.13	0.301	1.60	4.0
F02-EllipsoidalSep	droplet?	5.62	1.00	1.000	0.406	0.005	1.24	0.380	1.95	3.4
F03-RastriginSep	-	1.03	1.19	0.994	0.294	0.018	0.08	0.397	2.50	4.1
F04-BucheRastrigin	close	0.91	1.25	0.988	0.315	0.010	0.48	0.411	2.03	7.5
F05-LinearSlope	-	2.28	1.01	0.999	0.437	0.003	-0.09	0.290	1.71	4.2
Moderate — F06–09										
F06-AttractiveSector	close?	1.26	1.11	0.957	0.698	0.002	0.43	0.349	1.48	2.9
F07-StepEllipsoidal	-	1.57	1.05	0.781	0.944	0.009	-0.62	0.241	2.12	5.0
F08-Rosenbrock	-	2.56	1.01	0.897	0.986	0.007	-0.18	0.272	2.12	3.4
F09-RosenbrockRot	-	2.55	1.01	0.962	0.963	0.006	-0.07	0.293	2.00	3.4
Ill-conditioned — F10–14										
F10-EllipsoidalRot	droplet?	5.53	1.00	0.977	1.000	0.009	-0.28	0.243	1.96	3.1
F11-Discus	droplet	5.42	1.00	0.953	1.000	0.009	-0.26	0.239	1.99	3.2
F12-BentCigar	droplet	5.64	1.00	1.000	0.956	0.009	-0.29	0.271	2.09	3.8
F13-SharpRidge	droplet	4.89	1.00	0.956	1.000	0.005	0.13	0.257	1.67	6.0
F14-DiffPowers	droplet	3.34	1.00	0.965	0.993	0.002	-0.34	0.246	1.50	3.6
Multimodal — F15–19										
F15-RastriginRot	keep-air	1.07	1.18	0.861	0.774	0.018	0.01	0.343	2.33	4.4
F16-Weierstrass	close	1.63	1.05	0.994	0.486	0.050	0.21	0.427	3.35	4.1
F17-SchafferF7	keep-air	0.95	1.23	0.974	0.474	0.003	0.02	0.290	1.72	2.8
F18-SchafferF7III	?	2.63	1.00	0.974	0.978	0.008	-0.24	0.268	1.95	2.6
F19-GriewankRosenbrock	keep-air	0.73	1.38	0.951	0.513	0.101	-0.14	0.311	4.29	5.2

function	route	cond	PR	algA	offd	divs	kurt	mgap	nelt	spil
Weak structure — F20–24										
F20-Schwefel	keep-air	1.16	1.14	0.956	0.691	0.024	-0.27	0.313	2.38	6.5
F21-Gallagher101	-	1.31	1.10	0.799	0.897	0.017	-0.48	0.275	2.56	4.0
F22-Gallagher21	-	2.74	1.00	0.900	0.993	0.012	-0.56	0.239	2.29	4.1
F23-Katsuura	-	0.67	1.43	0.912	0.435	0.144	-0.09	0.282	5.49	3.4
F24-LunacekRastrigin	keep-air	0.64	1.47	0.876	0.498	0.076	0.21	0.325	4.31	13.1
Custom (multi-optimum) — C01–11										
C01-Himmelblau	-	0.83	1.30	0.933	0.547	0.001	1.32	0.603	1.81	2.4
C02-SixHumpCamel	-	1.28	1.21	0.980	0.609	0.053	-0.56	0.574	2.19	3.6
C03-Shubert	-	0.65	1.45	0.940	0.385	0.038	1.15	0.533	3.00	3.5
C04-FiveWell	-	0.46	1.62	0.915	0.309	0.002	-0.13	0.310	1.99	3.5
C05-Eggholder	keep-air	1.33	1.12	0.823	0.754	0.001	0.75	0.394	1.96	11.6
C06-Michalewicz	-	0.62	1.46	0.971	0.255	0.002	0.44	0.363	2.15	3.9
C07-BukinN6	-	4.15	1.00	0.993	0.992	0.001	1.26	0.335	1.67	13.0
C08-StyblinskiTang	-	0.49	1.59	0.935	0.313	0.001	0.79	0.415	1.65	3.9
C09-Easom	-	0.32	1.78	0.873	0.252	0.250	-0.73	0.252	5.30	7.6
C10-SchafferN2	-	0.34	1.76	0.894	0.258	0.031	-0.50	0.248	2.72	4.0
C11-DeJongF5	keep-air	0.54	1.53	0.951	0.300	0.006	-0.07	0.340	2.68	7.8